Mean-field theory of high-dimensional Bayesian inference, through the lens of spiked matrix estimation

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Goals

This course introduces statistical inference in high-dimensional settings, covering Bayesian foundations, information theory, challenges of dimensionality, before moving to more advanced concepts stemming from statistical mechanics and spin glasses, in particular mean-field methods. Prior knowledge of probability theory and linear algebra is assumed.

We will focus on a modern high-dimensional inference problem: the *spiked Wigner model*, a signal plus noise matrix model of the form

$$\mathbf{Y} = \tau\, \mathbf{x}\mathbf{x}^\intercal + \mathbf{Z} \in \mathbb{R}^{n\times n}.$$

It models in a mathematically simple manner the task of recovering a rank-one information matrix component $\mathbf{x}\mathbf{x}^\intercal$ (whose strength in the data is controlled by τ) hidden in a given large noisy data matrix $\mathbf{Y}$, which has been corrupted by Gaussian noise $\mathbf{Z}$. This is related to dimensionality reduction through *principal component analysis (PCA)*, which is probably one of the most basic problem in signal processing and machine learning. Despite our focus on this model, many of the methods and concepts provided in these notes extend to a much broader class of inference problems, see for instance these notes or this book to know more.

The first main question we will try to answer in a specific model is:

Can the signal $\mathbf{x}\mathbf{x}^\intercal$ be inferred from the data $\mathbf{Y}$? How well?

This really is an *theoretical/information-theoretic* question. Once the information-theoretic limits (i.e., the best possible performance) established we will then naturally ask ourselves:

Can it be inferred at low computational cost?

This question is an *engineering/algorithmic* question. These two questions complement each other: without a clear answer to the information-theoretic question, practitioners would lose time trying to improve algorithms with no hope of success, as they might already be close to optimal, or not at all, so that a completely different strategy must be designed. Indeed, the understanding of the barriers to information extraction provides guidance in algorithms design since the birth of information theory in 1948 with Claude Shannon.

An similar notes with additional material, rigorous justifications etc can be found here.

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Notations

Normal font x is for scalars, bold letters $\mathbf{x}$ are for vectors, bold capital letters $\mathbf{X} = (x_{ij})$ for matrices, capital letters X for generic variables of unspecified dimension (that can be scalars, vectors, matrices, etc).

Vectors $\mathbf{x} = (x_i)$ are columns, transposed vectors $\mathbf{x}^\intercal$ are rows. The inner product between vectors then reads $\mathbf{x}^\intercal\mathbf{y} = \sum_i x_i y_i$ also denoted as $\mathbf{x}\cdot\mathbf{y}$. For generic variables with same dimensions X, Y the inner product $X \cdot Y$ is defined similarly. E.g., for matrices, $\mathbf{X} \cdot \mathbf{Y} = \mathrm{Tr}\mathbf{X}\mathbf{Y}^\intercal$ where Tr is the trace.

The only norms we will use are $\|\mathbf{x}\| = (\sum_i x_i^2)^{1/2}$, the standard L_2 norm for vectors, and $\|\mathbf{X}\| = (\sum_{ij} x_{ij}^2)^{1/2}$, the Frobenius norm for matrices. For a generic variable/tensor X, $\|X\|$ is defined similarly with a sum over all its indices.

$\mathbb{E}_X, \mathbb{E}_{\mathbf{x}}$ etc are expectations w.r.t. to the random variables indicated by the subscript. $\mathbb{E}$ indicates an expectation w.r.t. *all* ensuing random variables appearing after it. In the Bayesian context, $\langle\cdot\rangle$ will be reserved for an expectation w.r.t. to the Bayesian posterior distribution $P(X|Y)$ if X denotes the unknown and $Y = Y(X)$ the observed data (the bracket $\langle\cdot\rangle$ will thus depend on the given data, even if not explicit from the notation). E.g., all the following notations are equivalent: $\mathbb{E}_{X|Y}[X] = \mathbb{E}[X|Y] = \langle X\rangle$. If multiple random samples $X = X^{(1)}, X^{(2)}, \ldots$ from the same posterior $P(\cdot|Y)$ appear inside an expression that includes an average over them, then the bracket acts jointly: $\langle g(X^{(1)}, X^{(2)}, \ldots)\rangle = \mathbb{E}[g(X^{(1)}, X^{(2)}, \ldots)|Y] = \int dP(X^{(1)}|Y) dP(X^{(2)}|Y) \cdots g(X^{(1)}, X^{(2)}, \ldots)$. The notation $X \sim P$ means that the probability distribution

of the random variable X is P. The notation $\mathcal{N}(m, \sigma^2)$ is the probability disribution of a Gaussian random variable with mean m and variance σ^2. Similarly, $\mathcal{N}(\mathbf{m}, \mathbf{C})$ is the multivariate density with given mean vector and covariance matrix.

With slight abuses of notations: *i*) we will *not* use different notations for a random variable and its outcomes/realizations, so whether a variable $X, \mathbf{x}, \dots$ is random or a fixed outcome depends on the context (e.g., if an expectation is present $\mathbb{E}_X$ then X is random and is averaged over all its outcomes). Following this, *ii*) we will denote $P(\mathbf{x}) = P_{\mathbf{x}}(\mathbf{x}), P(Y|X) = P_{Y|X}(Y|X), \dots$ where subscripts indicate to which random variable a law refers to; when no subscript is present, the argument will indicate so.

1 (Lect. 1) Bayesian inference 101, and the high-dimensional world

For the two first lectures, I highly recommend MacKay's book for those who want to go deeper.

1.1 Foundations of Bayesian inference

Bayesian inference provides a coherent probabilistic framework for updating beliefs about unknown quantities X (parameters, which can be a scalar, vector, tensor, etc) given observed data Y (which can have a different dimension). The core components are:

- **Prior distribution** $P(X)$: Represents our beliefs about X *before* seeing data
- **Likelihood** $P(Y|X)$: Probability of data given parameters. It models the data-generating process, parametrised by the unknown X
- **Posterior distribution** $P(X|Y)$: Updated beliefs about X *after* seeing the data
- **Evidence** $P(Y)$: The law of the data independently of the unknown

These components are connected through **Bayes' theorem**:

$$P(X|Y) = \frac{P(Y|X)P(X)}{\int P(Y|X)P(X)dX} = \frac{P(Y|X)P(X)}{P(Y)} \propto P(Y|X)P(X) \tag{1}$$

Example 1 (Coin tossing - discrete parameter)**.** *Suppose we want to estimate the mean $x \in [0, 1]$ of a possibly biased coin. We define:*

- ***Prior**: $x \sim Beta(2, 2), P(x) = x(1-x)$ (symmetric belief)*
- ***Data**: 3 heads in 5 tosses ($y = 3$, $n = 5$)*
- ***Likelihood**: Binomial $P(y|x) = \binom{5}{3} x^3(1-x)^2$*

The posterior is:

$$P(x|y) \propto x^3(1-x)^2 \cdot x(1-x) = x^{5-1}(1-x)^{4-1} \Rightarrow x|y \sim Beta(5, 4)$$

Posterior mean $= 5/(5+4) = 0.556$ (prior mean was 0.5)

Example 2 (Normal mean estimation - continuous parameter)**.** *Estimate average height $x = \mu$ in a population:*

- ***Prior**: $\mu \sim \mathcal{N}(\mu_0 = 170, \sigma_0^2 = 25)$ (mean 170cm, SD 5cm)*
- ***Data**: $\mathbf{y} = (\bar{\mu}, \bar{\sigma}^2)$ with $\bar{\mu} = 175cm$ from $n = 10$ samples, empirical $\bar{\sigma}^2 = 49cm$*

The posterior is Gaussian:

$$\mu|\mathbf{y} \sim \mathcal{N}\left(\frac{\bar{\sigma}^2\mu_0 + n\sigma_0^2\bar{\mu}}{n\sigma_0^2 + \bar{\sigma}^2}, \frac{\sigma_0^2\bar{\sigma}^2}{n\sigma_0^2 + \bar{\sigma}^2}\right) = \mathcal{N}\left(\frac{49 \cdot 170 + 10 \cdot 25 \cdot 175}{10 \cdot 25 + 49}, \frac{25 \cdot 49}{10 \cdot 25 + 49}\right) = \mathcal{N}(174.2, 4.1)$$

Data pulls the estimate from prior mean (170) toward the sample mean (175).

1.2 Bayesian decision theory

Once we have the posterior, how do we make optimal decisions? Decision theory formalizes this through:

Loss functions and risk

- **Loss function** $L(X, \delta)$: Cost of choosing decision δ when true parameter is X
- **Risk**: Expected loss for a decision rule $\delta(Y)$ given data Y:

$$R(Y, \delta) = \mathbb{E}_{X|Y}[L(X, \delta(Y))] = \mathbb{E}[L(X, \delta(Y))|Y] \tag{2}$$

- **Bayes risk**: Overall expected loss across all possible data Y:

$$r(\delta) = \mathbb{E}_Y[R(Y, \delta)] = \mathbb{E}[L(X, \delta(Y))] \tag{3}$$

Bayes-optimal estimators

The optimal decision minimizes the *risk, i.e., the posterior expected loss*:

$$\delta^*(Y) = \arg\min_{\delta} R(Y, \delta) \tag{4}$$

Example 3 (Mean-squared error loss)**.** *For L_2 loss (often used for real-valued variables) $L(X, \delta) = \|X - \delta\|^2$*

$$\begin{aligned} \mathbb{E}[\|X - \delta\|^2|Y] &= \mathbb{E}[\|X\|^2|Y] - 2\delta \cdot \mathbb{E}[X|Y] + \|\delta\|^2 \\ \nabla_\delta(\cdots) &= -2\mathbb{E}[X|Y] + 2\delta = 0 \\ \Rightarrow \delta^* = X_{\text{Bayes}} &= \mathbb{E}[X|Y] = \langle X \rangle \end{aligned}$$

Posterior mean = minimum mean-squared error (MMSE) estimator

Example 4 (0-1 loss for classification)**.** *For discrete-valued X and $L(X, \delta) = \mathbf{1}_{\{X \neq \delta\}}$*

$$\begin{aligned} \mathbb{E}[L(X, \delta)|Y] &= P(X \neq \delta|Y) = 1 - P(X = \delta|Y) \\ \Rightarrow \delta^* = X_{\text{MAP}} &= \arg\max_X P(X|Y) \end{aligned}$$

Posterior mode = maximum a posteriori (MAP) estimator

Why Bayesian framework in high dimensions?

$\mathbf{x} \in \mathbb{R}^p, \mathbf{y} \in \mathbb{R}^n$. In high-dimensional problems ($p \sim n$ or $p \gg n$):

- Maximum Likelihood Estimation (MLE) suffers from *overfitting* (e.g., unregularized regression fits noise)
- Bayesian approach automatically incorporates regularization through the prior
- Example: Ridge regression as posterior mean under Gaussian prior

$$\begin{aligned} \mathbf{y} = \mathbf{\Phi}\mathbf{x} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 \boldsymbol{I}) \\ P(\mathbf{x}) \sim \mathcal{N}(0, \tau^2 \boldsymbol{I}) \\ \Rightarrow \mathbf{x}_{\text{Bayes}} = \langle \mathbf{x} \rangle = (\mathbf{\Phi}^\intercal \mathbf{\Phi} + \lambda \boldsymbol{I})^{-1} \mathbf{\Phi}^\intercal \mathbf{y} \quad (\lambda = \sigma^2/\tau^2) \end{aligned}$$

  Prior effectively "shrinks" estimates toward zero, reducing variance

1.3 The curse of dimensionality

The **curse of dimensionality** refers to phenomena that arise when analyzing data in high-dimensional spaces. As dimensionality increases, data becomes sparse, distances lose meaning, and naive inference and learning algorithms degrade.

Table 1: Common loss functions and Bayes estimators

Loss function	$L(X, \delta)$	Bayes estimator
Squared error	$\|X - \delta\|^2$	Posterior mean (MMSE)
Absolute error	$\|X - \delta\|$	Posterior median
0-1 loss	$\mathbf{1}_{\{X \neq \delta\}}$	Posterior mode (MAP)

Exponential sample requirement

To maintain constant density in $[0,1]^d$:

1D: n points cover $\Delta x = 1/n$

dD: n points per dimension to maintain the same resolution (i.e., distance between neighbor points)
For $d = 100,\ 100^{10} = 10^{20}$ points needed for 0.01-resolution

Real-world datasets are always sparse in high dimensions.

Distance concentration

For $\mathbf{x}, \mathbf{y}$ sampled uniformly in $[0,1]^d$:

$$\mathbb{E}[\|\mathbf{x} - \mathbf{y}\|^2] = \frac{d}{6}, \quad \text{Std}(\|\mathbf{x} - \mathbf{y}\|^2) = \sqrt{\frac{7d}{180}} \quad \Rightarrow \quad \frac{\text{dist}^2_{\max} - \text{dist}^2_{\min}}{\text{dist}^2_{\min}} = O(d^{-1/2}) \to 0 \text{ as } d \to \infty$$

All distances become similar in high dimensions.

Boundary dominance

For d-sphere with radius r:

$$\frac{\text{Vol}([r-\epsilon, r])}{\text{Vol}([0, r])} = 1 - \left(1 - \frac{\epsilon}{r}\right)^d \xrightarrow{d \to \infty} 1$$

Most volume concentrates near the boundary.

Example 5 (Classification Example: k-NN Failure)**.** *Problem Setup:*

- ***Space****:* $[0,1]^d$ *hypercube*
- ***Training data****:* $n = 1000$ *points (sparse)*
- ***Decision boundary****:* $\sum_{i=1}^d x_i = d/2$*. I.e, label* $y(\mathbf{x}) = 1(\sum_{i=1}^d x_i \geq d/2)$*.*
- ***Classifier****: 1-Nearest Neighbor (1-NN)*

	$d = 2$	$d = 200$	$d = 2000$
Mean distance between points	0.52	5.8	18.3
Std of distance	0.25	0.3	0.63
Mean nearest neighbor distance	0.015	5	17.5

Table 2: How distances between uniformly sampled points evolve with the dimension. Data obtained from simulations (no simple formula).

From Table 2 we see that for a uniformly sampled test point, the probability that 1-NN fails increases rapidly with d: most points, including the closest neighbour of any point, lie in a thinner shell as d increases, and the fluctuations among distances become too large compared to the nearest neighbour distance to a point to classify reliably.

The curse of dimensionality fundamentally changes data geometry:

- Distances become uninformative
- Data concentrates in boundary regions
- Local neighbourhoods cease to exist

Understanding these phenomena is crucial for effective high-dimensional data analysis. The curse can be mitigated by incorporating structural assumptions about the data.

Solutions to combat the curse: **dimensionality reduction** (e.g., PCA), or similarly, **selection of few important features** (via mutual information, LASSO, etc), **low-dimensional manifold assumptions**, i.e., assume data lies on low-dimensional manifold (to be found), and **regularization** to prevent overfitting to sparse data.

All these techniques can be seen as different forms of priors: without making a-priori assumptions, inference in high-dimensional spaces is doomed. Bayesian inference is thus very important in these settings.

2 (Lect. 2) Information theory 101: "Hi, my name is Claude"

We want to precisely quantify when statistical inference is possible or not by estimating if the data contains enough information about the model parameters. But what is information, and how to quantify it? The answer has been understood by Claude Shannon in 1948 in a seminal paper which started information theory: the right notion is the entropy. Computing entropies and relatives will be one of our main goal, so let us start by understand why it is the proper definition of information content conveyed by a random variable.

Warning

Remember that, in contrast with the standard notations of information theory, we are *not* distinguishing notation-wise between a random variable X and its outcomes. The meaning should be clear from the context.

2.1 Surprise, Shannon entropy and its properties

For simplicity let us consider a discrete setting. Let $X \sim P$ a discrete r.v. with possible outcomes in $\mathcal{X} = \{X\}$. Recall that X can be any random object. The *Shannon entropy* $H(X) = H(P)$ of the random variable X is defined as:

$$H(X) = H(P) := \sum_{X \in \mathcal{X}} P(X) \ln \frac{1}{P(X)} = \sum_{i=1}^{|\mathcal{X}|} P_i \ln \frac{1}{P_i}. \tag{5}$$

It is the expectation with respect to P of the *information content*, or *surprise*,

$$h(X) = h(P(X)) := \ln \frac{1}{P(X)}.$$

If an outcome has low probability then observing it is quite surprising, and it brings a lot of information as it was not expected: its surprise is high. If instead $P(X)$ is close to 1 it is not surprising to observe outcome X, so this outcome brings low information: $h(X)$ is low. Said differently: if the outcome of a random variable is very probable, it is no surprise (and generally uninteresting) when it happens, because it was expected. However, if an outcome is unlikely to occur, it is much more informative if it happens to be observed. The term information content must be understood as a *potential* information gain if the outcome was to be observed. Here the information content and entropy are expressed in "nats" (for "natural units"), because the logarithm is in natural basis. When using the $\log_2$ they are expressed in "bits".

Imagine you are in the desert and suddenly it rains like hell. Worst, it rains cows that play piano! What? It is amazingly surprising no? The probability of this event is actually so low that it brings an enormous amount of information; in this case it should lead you to the conclusion that you are dreaming. If instead

your are in the desert and its super sunny and hot, it is not surprising at all; this does not bring more information than what you already know, and if you are dreaming, it is unlikely that this observation will help you realize it. Another example: the knowledge that some particular number will not be the winning one of a lottery provides very little information, because any particular chosen number will almost certainly not win. However, knowledge that a particular number will win a lottery has high informational value because it communicates the outcome of a very low probability event.

Be focused here. The entropy can also be interpreted as a *measure of unpredictability* of X, or of *uninformation/lack of knowledge* about what X's outcome will be: the more surprising are the outcomes in average, the more unpredictable is the actual outcome, which also mean the less we know about X *before* observing it. $H(X)$ *quantifies the amount of missing information necessary to determine the outcome of* X *before observing it.* This can be confusing because previously we said that $H(X)$ is an expected information content, while now we speak about a measure of uninformation. There is no paradox: an information $H(X)$ is *gained* in expectation when X *is actually observed.* But *before* observing it, $H(X)$ measures the uninformation about it. Put differently: observing X *converts* in average $H(X)$ units of uninformation into information. So it just a matter of conceptually placing ourselves *before* X is observed –in which case the interpretation as a measure of uninformation may be more natural–, or *after* X is observed –where the interpretation as an expected information content fits better.

An example might help: the outcome of a toss of a fair coin $x_{\text{fair}} \sim \text{Ber}(1/2)$ is much more unpredictable than the outcome of a strongly biased coin $x_{\text{bias}} \sim \text{Ber}(9/10)$, or equivalently our lack of knowledge about x_{fair} is higher: we are more uninformed. But when *observing* the outcome of the fair coin, we then *gain* more information than with the unfair one, because it is in average more surprising. In the first case, which has entropy $H(x_{\text{fair}})$ of one bit, betting on one side or the other is the same statistically. While in the second case, where $H(x_{\text{bias}}) = \frac{9}{10}\log_2\frac{10}{9} + \frac{1}{10}\log_2 10 \approx 0.47$, the outcome is much more predictable, we are less uninformed (= more informed); it would be an error not to bet on the outcome $x_{\text{bias}} = 1$.

The Shannon entropy $H(X)$ of the random variable X (or equivalently of its law) quantifies:

i) Its average information content, i.e., the expected information gain when observing the outcome of X.

ii) The average uninformation/lack of knowledge about the outcome prior to observe it.

iii) Its unpredictability: the higher the entropy of X, the less "structured" its distribution is.

Finally, if expressed in bits, $H(X)$ is the expected number of binary "yes/no" questions required to determine the outcome *before* it is observed, or equivalently, the expected number of binary questions that the outcome of X answered *after* being observed.

Properties that make Shannon entropy a good measure of information content

Let us provide some properties suggesting that $H(X)$ is *the* proper definition of information content of X (the formal result by Claude Shannon showing that it indeed is the proper notion of information is called the source coding theorem). Let $X \sim P$.

- $H(X) \geq 0$ with equality if and only if $P_i = 1$ for one i. There is no such thing as negative information, and a deterministic variable convey no information.
- $H(X)$ is maximized if P is uniform: $H(X) \leq \ln|\mathcal{X}|$ with equality if and only if $P_i = \frac{1}{|\mathcal{X}|}$ for all i.

So the uniform distribution (the less structured of them all) has maximum entropy, while the trivial distribution giving full weight to a single event has 0 entropy. As the entropy of the uniform distribution increases with $|\mathcal{X}|$, casting a die has higher entropy than tossing a coin because each outcome of a die toss has smaller probability (1/6 to be compared to the 1/2 of the fair coin; each outcome of a die is more surprising as there are more of them).

Notice another nice property of the information content function $h(x) = -\ln x$. Imagine learning the outcome x and y of two independent random variables, X and Y. Intuitively, we might want any measure of the "amount of information gained" to have the property of additivity: for independent random variables X and Y, the information gained when we learn the outcome of both should equal the sum of the information gained from both individually. And indeed, $h(X,Y) = h(X) + h(Y)$ if $X \perp Y$ (i.e., are independent: $P_{X,Y} = P_X \otimes P_Y$). Consequently,

- $P_{X,Y} = P_X \otimes P_Y \;\Rightarrow\; H(X,Y) = H(X) + H(Y)$.

Here

$$H(X,Y) = H(P_{X,Y}) := \sum_{X,Y \in \mathcal{X}\times\mathcal{Y}} P(X,Y) \ln \frac{1}{P(X,Y)}$$

is the entropy of the joint distribution, $h(X,Y) = h(P(X,Y)) = -\ln P(X,Y)$. In words, *entropy is additive for independent variables*. Define also the *conditional entropy* of X given Y:

$$H(X|Y) := \sum_{X,Y \in \mathcal{X}\times\mathcal{Y}} P(Y)P(X|Y) \ln \frac{1}{P(X|Y)}.$$

It is equal to the expected information revealed by observing X *given* the outcome of Y was already observed. Or equivalently, it is the remaining amount of unpredictability of X given the observation of Y.

- The entropy $H(X,Y)$, i.e., the amount of information revealed by simultaneously evaluating (X,Y) equals the information revealed by conducting two consecutive experiments: first evaluating the value of Y, then revealing the value of X *given* that you know the value of Y (or the other way around). This is called the *chain rule for entropy*:

  $$H(X,Y) = H(X|Y) + H(Y) = H(Y|X) + H(X).$$

- If f is a function then $H(f(X)|X) = 0$. Applying that to the previous formula yields $H(X) + H(f(X)|X) = H(f(X)) + H(X|f(X))$, so:

  $$H(f(X)) \leq H(X),$$

  with equality if f is invertible. The entropy may only decrease when X is passed through a function.

- If X and Y are independent then knowing the value of Y doesn't influence our knowledge of the value of X (since the two don't influence each other by independence):

  $$P_{X,Y} = P_X \otimes P_Y \;\Rightarrow H(X|Y) = H(X),\; H(Y|X) = H(Y).$$

- The entropy of two simultaneous events is no more than the sum of the entropies of each individual event, and are equal if the two events are independent:

  $$H(X,Y) \leq H(X) + H(Y).$$

- The entropy seen as a function of the probability distribution P is concave: for all $P^{(1)}$ and $P^{(2)}$ and $\lambda \in [0,1]$,

  $$H(\lambda P^{(1)} + (1-\lambda)P^{(2)}) \geq \lambda H(P^{(1)}) + (1-\lambda)H(P^{(2)}).$$

  Note that $\lambda P^{(1)} + (1-\lambda)P^{(2)}$ is a probability distribution.

2.2 Kullback-Leibler divergence: measuring distribution mismatch

First let us generalise the definition of entropy to continuous X with density $f(x)$.

Definition 1 (Differential entropy)**.** *The differential entropy reads*

$$\mathsf{h}(X) = -\int_{\mathcal{X}} f(X) \log f(X) dX \quad \textit{(nats)} \tag{6}$$

Be careful, the differential entropy can be negative, so the interpretations in terms of information content etc. holding for the entropy of discrete variables are lost. The correct object to talk about information in general, including with continuous variables, is the mutual information defined soon. Abusing notation, we will use in the remaining the same notation H as for Shannon entropy also for the differential entropy of continuous variables, as must ensuing expressions apply in both cases.

Definition 2 (KL divergence). *For distributions P and Q on $\mathcal{X}$:*

$$D_{KL}(P\|Q) = \sum_X P(X) \log_2 \frac{P(X)}{Q(X)} \quad \textit{(discrete)}$$

$$D_{KL}(P\|Q) = \int f_P(X) \log_2 \frac{f_P(X)}{f_Q(X)} dX \quad \textit{(continuous)}$$

- Measures "inefficiency" of assuming Q when true distribution is P
- Expected extra bits needed to code samples from P using an optimal compression code for samples from Q.

Example 6 (Gaussian distributions). *Let $P = \mathcal{N}(\mu_1, \sigma^2)$, $Q = \mathcal{N}(\mu_2, \sigma^2)$:*

$$D_{KL}(P\|Q) = \frac{(\mu_1 - \mu_2)^2}{2\sigma^2}$$

The divergence grows with mean separation and decreases with the variance.

KL divergence properties

- **Non-negativity**: $D_{\mathrm{KL}}(P\|Q) \geq 0$ with equality iff $P = Q$
- **Asymmetry** (*not* a true distance): $D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P)$
- **Additivity**: $D_{\mathrm{KL}}(P_X \otimes P_Y \| Q_X \otimes Q_Y) = D_{\mathrm{KL}}(P_X\|Q_X) + D_{\mathrm{KL}}(P_Y\|Q_Y)$
- **Data processing**: If Y is function of X, $D_{\mathrm{KL}}(P_X\|Q_X) \geq D_{\mathrm{KL}}(P_Y\|Q_Y)$
- **Connection to MLE**: Given a model $P_{Y|X}$ depending on unknown parameters X for the i.i.d. data $(Y_1, Y_2, \ldots, Y_n)$, whose true distribution for each independent sample is $\bar{P}_Y$, maximizing the log-likelihood of the observed data according to the model is asymptotically equivalent to minimizing the KL divergence between the model and the true data distribution $\bar{P}_Y$:

$$\arg\max_X \sum_{i \leq n} \log P_{Y|X}(Y_i|X) \to \arg\min_X D_{\mathrm{KL}}(\bar{P}_Y \| P_{Y|X}) \quad \text{as} \quad n \to \infty$$

2.3 Mutual information: Measuring statistical dependence

Definition 3 (Mutual information). *For a pair (X, Y) of random variables the mutual information reads*

$$I(X;Y) = I(Y;X) = D_{KL}(P_{X,Y}\|P_X \otimes P_Y) = \sum_{X,Y \in \mathcal{X}\times\mathcal{Y}} P(X,Y) \log_2 \frac{P(X,Y)}{P(X)P(Y)}$$

For the case of continuous random variables, summation becomes integration and the probabilities are replaced by densities.

Equivalent expressions, if interpreting X as signal and Y as data:

$$I(X;Y) = H(X) - H(X|Y)$$

Differential entropy should be used instead of the Shannon entropy for continuous variables. The mutual information is thus the uncertainty about the signal minus the remaining uncertainty about the signal when the data is observed (i.e., due to noise) = reduction of uncertainty about X when observing Y = information carried by the data about the signal.

$$I(X;Y) = H(Y) - H(Y|X)$$

This is also the total information carried but the data minus the remaining information contained in the data when the signal is already known (i.e., contribution due to noise) = information carried by the data about the signal.

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

Given a third random variable, the conditional mutual information is defined similarly

$$I(X;Y|Z) = I(Y;X|Z) = D_{KL}(P_{X,Y|Z} \| P_{X|Z} \otimes P_{Y|Z}) = \sum_{X,Y,Z} P(X,Y,Z) \log_2 \frac{P(X,Y|Z)}{P(X|Z)P(Y|Z)}$$
$$= H(X|Z) - H(X|Y,Z) = H(Y|Z) - H(X|Y,Z)$$

and similarly for continuous variables with densities, with integrals replacing sums, and the differential entropy replacing the Shannon entropy.

Key properties of mutual information

- **Symmetry**: $I(X;Y) = I(Y;X)$
- **Non-negativity**: $I(X;Y) \geq 0$ with equality iff $X \perp Y$
- **Data Processing Inequality**: Markov chain $X \to Y \to Z \Rightarrow I(X;Y) \geq I(X;Z)$
- **Chain Rule**: $I(X;Y,Z) = I(X;Y) + I(X;Z|Y)$
- **Entropy bound (for discrete variables only)**: $I(X;Y) \leq \min(H(X), H(Y))$

Example 7 (Noisy channel)**.** *Consider a binary symmetric channel with flip probability ϵ:*

- *Input $X \sim$ Bernoulli(p), output $Y = X \oplus Z$ with $Z \sim$ Bernoulli(ϵ)*
- *Mutual information: $I(X;Y) = H(Y) - H(Y|X) = H(X) - H(X|Y)$*
- *Maximum at $\epsilon = 0$: $I(X;Y) = H(X) = H_b(p)$ (the binary entropy)*
- *Minimum at $\epsilon = 0.5$: $I(X;Y) = 0$ (no information)*

2.4 I-MMSE formula: Connecting information to estimation

The I-MMSE formula establishes a fundamental link between information theory and estimation theory in Gaussian channels.

Theorem 1 (I-MMSE formula)**.** *Let Z be made of i.i.d. standard Gaussian entries drawn from $\mathcal{N}(0,1)$. Consider the Gaussian channel:*

$$Y = \sqrt{\lambda} X + Z$$

where X is the input signal (scalar, vector, matrix, etc), Y is the observation, and $\lambda \geq 0$ is signal-to-noise ratio (SNR). Then:

$$\frac{d}{d\lambda} I(X;Y) = \frac{1}{2} \mathrm{MMSE}(X|Y) = \frac{1}{2} \mathbb{E} \|X - \mathbb{E}[X|Y]\|^2,$$

where the latter is the minimum mean-squared error (MMSE), i.e., the lowest achievable (in average) mean-squared estimation error on the signal.

We will prove the I-MMSE formula in the next lecture.

Interpretation:

- Derivative of mutual information equals half the MMSE
- As SNR (γ) increases, mutual information increases, optimal estimation error decreases
- Fundamental connection between how much we know (information) and how well we can estimate

3 (Lect. 3) The main character: Spiked Wigner model, and Bayesian-optimal performance

3.1 Bayesian-optimal setting and the Nishimori identity

Notice the following important property of Bayesian-optimal inference, also called "Nishimori identity" in physics, which is simply a consequence of the tower property of conditional expectation

$$\mathbb{E}_{X,Y}[\,\cdot\,] = \mathbb{E}_X[\mathbb{E}_{Y|X}[\,\cdot\,]] = \mathbb{E}_Y[\mathbb{E}_{X|Y}[\,\cdot\,]].$$

Theorem 2 (Nishimori identity for Bayes-optimal inference)**.** *Let $X, Y \sim P_{X,Y}$. Let $X^{(1)}, \dots, X^{(k)}$ be i.i.d. random variables with distribution $P_{X|Y}$. Let $\mathbb{E}$ be the expectation w.r.t. $P_{X,Y}$ and $\langle - \rangle$ w.r.t. the product measure $P_{X|Y}^{\otimes\infty}$. Then, for any continuous bounded function g,*

$$\mathbb{E}\big\langle g(Y, X, X^{(2)}, \dots, X^{(k)})\big\rangle = \mathbb{E}\big\langle g(Y, X^{(1)}, X^{(2)}, \dots, X^{(k)})\big\rangle. \tag{7}$$

Proof. This directly follows from Bayes formula $P_{XY} = P_{X|Y}P_Y = P_{Y|X}P_X$. It is equivalent to sample the couple X, Y according to its joint distribution or to sample first Y according to its marginal distribution and then to sample X conditionally on Y from the conditional distribution. Thus $(Y, X, X^{(2)}, \dots, X^{(k)})$ and $(Y, X^{(1)}, X^{(2)}, \dots, X^{(k)})$ have the same law. □

Example: Various expressions for the MMSE

This identity is at the origin of many simplifications that allow to carry a complete analysis of high-dimensional inference problems in the *Bayesian optimal setting*. In an inference setting this means that the posterior $P_{X|Y}$ is known and correctly exploited by the statistician. Any other setting is called *mismatched*, e.g. when the prior used by the statistician Q_X does not match the true one P_X that generated the signal, or the data-generating model given by the likelihood used $Q_{Y|X} \neq P_{Y|X}$, etc. The Nishimori identity only applies in the Bayes-optimal setting as can be understood from its proof.

The Nishimori implies for instance that

$$\mathrm{MMSE}(X|Y) = \mathbb{E}\big\langle \|X - \langle X\rangle\|^2\big\rangle = \mathbb{E}\|X^* - \langle X\rangle\|^2,$$

where $X \sim P(\,\cdot\,|Y)$ while $X^* \sim P_x$ (from now on, we will often emphasize with a superscript $*$ a random variable drawn from the prior P_X in order to distinguish it from samples from the posterior $P(\,\cdot\,|Y)$). Indeed, $\mathbb{E}\langle\|X - \langle X\rangle\|^2\rangle$ is just a function of $X \sim P(\,\cdot\,|Y)$ and of $\langle X\rangle$ which only depends on Y, so by the Nishimori identity X can be replaced by the signal X^*. Another simplification thanks to it is

$$\mathbb{E}\big\langle \|X - \langle X\rangle\|^2\big\rangle = \mathbb{E}\big\langle \|X\|^2\big\rangle - \mathbb{E}\|\langle X\rangle\|^2 = \mathbb{E}\|X^*\|^2 - \mathbb{E}\big\langle X^{(1)} \cdot X^{(2)}\big\rangle = \mathbb{E}\|X^*\|^2 - \mathbb{E}\langle Q\rangle, \tag{8}$$

where the *overlap* between a posterior sample and the signal is

$$Q(X, X^*) = Q := X^* \cdot X.$$

Variables $X^{(1)} = X$ and $X^{(2)}$ are i.i.d. samples from the posterior $P(\,\cdot\,|Y)$, called *replicas*. We used, again thanks to the Nishimori identity, that

$$\mathbb{E}\|\langle X\rangle\|^2 = \mathbb{E}\big\langle X^{(1)} \cdot X^{(2)}\big\rangle = \mathbb{E}\big[X^* \cdot \langle X\rangle\big].$$

3.2 The scalar denoising problem

We start with the simplest possible inference problem: denoising a scalar parameter corrupted by Gaussian noise, i.e., the data

$$\text{Scalar denoising problem:} \quad y = \sqrt{\lambda}x^* + z,$$

with $z \sim \mathcal{N}(0, 1)$, with prior $x^* \sim P_x$. In general, computing the mutual information $I(x^*; y)$ and the minimum mean-square error $\mathrm{MMSE}(x^*|y) = \mathbb{E}[(x^* - \langle x\rangle)^2] = \mathbb{E}\langle(x - \langle x\rangle)^2\rangle$ requires a computer. One case where it is explicit is for Gaussian prior. Again, the starred variable x^* is the signal (drawn from the prior P_x), while $x = x|y$ is for a sample from the posterior $P_{x|y}$.

Example 8 (Gaussian signal)**.** *If* $x^* \sim P_x = \mathcal{N}(0, \sigma^2)$,

$$I(x^*; y) = \frac{1}{2}\ln(1+\lambda\sigma^2), \quad \text{MMSE}(x^*|y) = \frac{\sigma^2}{1+\lambda\sigma^2}.$$

I-MMSE relation: $2\dfrac{d}{d\lambda} I(x^*; y) = \dfrac{\sigma^2}{1+\lambda\sigma^2} = \text{MMSE}(x^*|y).$

In general, the mutual information for the denoising problem reads

$$\begin{aligned}
I(x; y) &= H(y) - H(z) \\
&= -\int dy dP_x(x^*) \frac{e^{-\frac{1}{2}(y-\sqrt{\lambda}x^*)^2}}{\sqrt{2\pi}} \ln \int dP_x(x) \frac{e^{-\frac{1}{2}(y-\sqrt{\lambda}x)^2}}{\sqrt{2\pi}} - \frac{\ln(2\pi e)}{2} \\
&= -\int dz dP_x(x^*) \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}} \ln \int dP_x(x) \frac{e^{-\frac{1}{2}(\sqrt{\lambda}x^*+z-\sqrt{\lambda}x)^2}}{\sqrt{2\pi}} - \frac{\ln(2\pi e)}{2} \\
&= -\mathbb{E}_{z\sim\mathcal{N}(0,1)}\mathbb{E}_{x^*\sim P_x} \ln \int dP_x(x) e^{\lambda x^* x + \sqrt{\lambda} z x - \frac{\lambda}{2}x^2} + \frac{1}{2}\mathbb{E}[(\sqrt{\lambda}x^* + z)^2] - \frac{1}{2},
\end{aligned}$$

which finally gives

$$I(x^*; y) = \frac{\lambda \mathbb{E}[x^{*2}]}{2} - \mathbb{E}_{z,x^*} \ln \int dP_x(x) e^{\lambda x^* x + \sqrt{\lambda} z x - \frac{\lambda}{2}x^2}. \tag{9}$$

Let us prove the I-MMSE formula for the scalar denoising problem; the generalisation beyond the scalar case is trivial and only requires keeping track of more indices. Consider the following derivative of the mutual information:

$$2\frac{d}{d\lambda} I(x^*; y) = \mathbb{E}[x^{*2}] - 2\mathbb{E}\Big\langle \Big(x^* x + \frac{1}{2\sqrt{\lambda}} z x - \frac{1}{2}x^2 \Big) \Big\rangle \tag{10}$$

where the posterior average $\langle \cdot \rangle$ is w.r.t.

$$P(x|y(x^*, z)) = \frac{P_x(x) e^{\lambda x^* x + \sqrt{\lambda} z x - \frac{\lambda}{2}x^2}}{\int dP_x(x) e^{\lambda x^* x + \sqrt{\lambda} z x - \frac{\lambda}{2}x^2}} \propto P_x(x) e^{-\frac{1}{2}(y-\sqrt{\lambda}x)^2}. \tag{11}$$

Now we can integrate by parts the Gaussian variable z using the following simple formula (sometimes called Stein lemma): for a function f with not increasing too fast at $\pm\infty$ (i.e., such that $\lim_{t\to\pm\infty} f(t)e^{-t^2/2} = 0$),

$$\mathbb{E}[z f(z)] = \mathbb{E}[f'(z)]. \tag{12}$$

Here the function of z is the posterior average $\langle x \rangle(y)$ that depends on it through the data $y = y(x^*, z)$. Using this formula gives

$$2\frac{d}{d\lambda} I(x^*; y) = \mathbb{E}[x^{*2}] - 2\mathbb{E}\Big\langle \Big(x^* x - \frac{1}{2}x^2 \Big) \Big\rangle - \mathbb{E}[\langle x^2 \rangle - \langle x \rangle^2]. \tag{13}$$

You should be able to show it, being careful that also the normalisation in the posterior (11) depends on the same z. A direct application of the Nishimori identity then gives (via $\mathbb{E}[x^* \langle x \rangle] = \mathbb{E}[\langle x \rangle^2]$)

$$\cdots = \mathbb{E}[x^{*2}] - 2\mathbb{E}[x^* \langle x \rangle] + \mathbb{E}[\langle x \rangle^2] = \mathbb{E}[x^{*2}] - \mathbb{E}[\langle x \rangle^2] = \mathbb{E}[x^{*2}] - \mathbb{E}[x^* \langle x \rangle] \tag{14}$$

which is also equal, again by the Nishimori identity, to

$$\cdots = \mathbb{E}[\langle x^2 \rangle - \langle x \rangle^2] = \mathbb{E}\langle (x - \langle x \rangle)^2 \rangle = \text{MMSE}(x^*|y). \tag{15}$$

We have proven the I-MMSE formula. Using the last form in (14) the MMSE explicitly reads

$$\text{MMSE}(x^*|y) = \mathbb{E}[x^{*2}] - \mathbb{E}_{x^*,z}\Big[x^* \frac{\int dP_x(x)\, x\, e^{\lambda x^* x + \sqrt{\lambda} z x - \frac{\lambda}{2}x^2}}{\int dP_x(x) e^{\lambda x^* x + \sqrt{\lambda} z x - \frac{\lambda}{2}x^2}} \Big]. \tag{16}$$

In the case of Gaussian prior, the integrals can be solved explicitly and it gives the expression in the last example. You should try showing it. Another case you can try to work out is that of Rademacher prior.

Example 9 (Rademacher signal)**.** *For* $P_x = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_{-1}$ *the MMSE function reads*

$$\mathrm{MMSE}(x^*|y) = 1 - \mathbb{E}_z[\tanh(\lambda + \sqrt{\lambda}z)^2] = 1 - \mathbb{E}_z[\tanh(\lambda + \sqrt{\lambda}z)]. \tag{17}$$

There is no typo here: with or without the square, these two expressions are the same as a consequence of the Nishimori identity. Let's plot this for Rademacher and standard Gaussian prior P_x with unit variance.

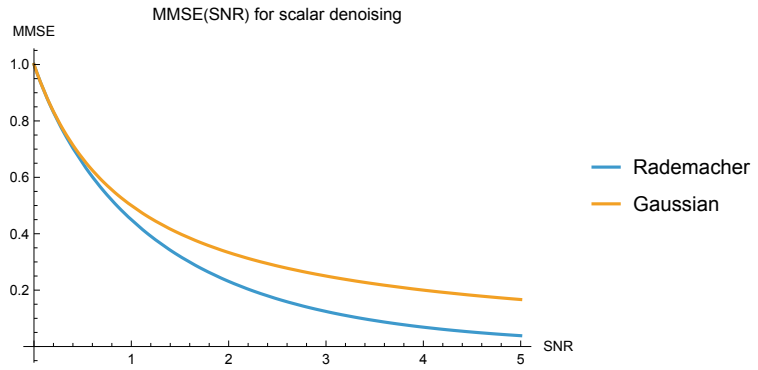


Figure 1: $\mathrm{MMSE}(x^*|\sqrt{\lambda}x^* + z)$ as a function of λ for the scalar denoising problem, for two choices of P_x.

The plot shows three things.

- The MMSE is 0 as $\lambda = 0$: the best one can do without data is to use the prior. The Bayes estimator is thus the prior mean (0) and its average error is the prior second moment: this is seen from (16) or direcly as
$$\mathrm{MMSE}(x^*|y) = \mathbb{E}[\langle x^2\rangle - \langle x\rangle^2] \le \mathbb{E}\langle x^2\rangle = \mathbb{E}[x^{*2}] \ (= 1)$$
using Nishimori.
- The Rademacher prior is more informative than the Gaussian one (i.e., constrain more the signal), and thus the corresponding MMSE stays below.
- The curve are smoothly decreasing (and are boring): there are no cusps, discontinuities, or other non-analytic points (i.e., phase transitions).

The reason why these curves are analytic and boring is that the problem is too simple. To make it more interesting, we need to introduce more complexity by considering *many variables* that *interact*. Only in this case we have a chance to observe interesting phase transition phenomena. This where the spiked Wigner model enters. However, studying the denoising problem was important because we will see that, surprisingly, it will be deeply connected to more complicated inference models like the spiked Wigner model through the "mean-field analysis".

3.3 The spiked Wigner model

The spiked Wigner model is a simple probabilistic model of principal component analysis. This model is rich enough so that it contains all features of more complex inference problems (but not too much like the dnoising problem), and remains simple enough to fully analyse it.

We consider a signal-vector $\boldsymbol{x}^* = (x_i^*)_{i=1}^n$. Its entries x_i^* are drawn i.i.d. from P_x with second moment $\rho = \mathbb{E}[x_i^{*2}]$. The symmetric data-matrix $\mathbf{Y} = (y_{ij})_{i,j=1}^n$ is obtained through the following generative process, or "observation model":

$$\text{Spiked Wigner model:} \quad y_{ij} = \sqrt{\frac{\lambda}{n}} x_i^* x_j^* + z_{ij}, \quad 1 \le i < j \le n, \tag{18}$$

where the noise-matrix enries z_{ij} are i.i.d. Gausian $\mathcal{N}(0,1)$ for $i < j$, with $z_{ij} = z_{ji}$. The inference task is to estimate the rank-one "spike" $(x_i^* x_j^*)$ from the data (y_{ij}) corrupted by a Wigner noise matrix (z_{ij}). The

scaling $1/\sqrt{n}$ in the observation model (18) is here to make the spectral norm (amplitude of the maximum singular value) of the rank-one and noise matrix comparable, so that non-trivial estimation is possible. Any other scaling would make the estimation task either trivial or impossible. The model can also be written in matrix form as

$$\text{Spiked Wigner model:} \quad \mathbf{Y} = \sqrt{\frac{\lambda}{n}}\mathbf{x}^* \otimes \mathbf{x}^* + \mathbf{Z}, \tag{19}$$

where $\mathbf{x}^* \otimes \mathbf{x}^* = \mathbf{x}^*\mathbf{x}^{*\intercal} \in \mathbb{R}^{n\times n}$ is the rank-one spike. The only difference is the diagonal that we now observe. But it can be shown easily that in the large n limit the diagonal observations do not influence the problem (because the information carried by the diagonal $(y_{ii}^2)_{i\le n}$ is only n numbers, which is vanishingly small compared to information of the $n(n-1) = O(n^2)$ off-diagonal elements).

From Bayes' formula the posterior distribution reads

$$P(\boldsymbol{x}|\boldsymbol{Y}) \propto \prod_{i=1}^{n} P_X(x_i) \times \frac{1}{(2\pi)^{\frac{n(n-1)}{4}}} \exp\Big\{ -\frac{1}{2} \sum_{1\le i<j\le n} \Big(y_{ij} - \sqrt{\frac{\lambda}{n}} x_i x_j\Big)^2 \Big\}. \tag{20}$$

Applications of the Wigner spiked model

- *Sparse PCA:* In the simplest case the prior $P_x = \mathrm{Ber}(\rho)$. The task is to estimate the hidden sparse, low-rank representation $\mathbf{x}^* \otimes \mathbf{x}^*$ (and then $\pm\mathbf{x}^*$ by eigenvalue decomposition) of $\mathbf{Y}$.
- *Submatrix localization:* Again $P_x = \mathrm{Ber}(\rho)$. One has then to extract a submatrix of $\mathbf{Y}$ of size $\approx \rho n \times \rho n$ with larger mean.
- *Community detection in the stochastic block model:* Recovering two communities of size ρn and $(1-\rho)n$ in a dense SBM of n vertices is "equivalent" to the spiked Wigner model with

$$P_x = \rho\delta_{\sqrt{(1-\rho)/\rho}} + (1-\rho)\delta_{-\sqrt{\rho/(1-\rho)}}. \tag{21}$$

- $\mathbb{Z}/2$ *synchronization:* The prior is Rademacher $P_x = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$. The task is to determine the nodes states $\mathbf{x}^* \in \{-1,1\}^n$ (up to a global sign) from noisy obervations of the "agreements" $x_i^* x_j^*$.

There exists a non-symmetric version called *spiked Wishart model*:

$$\text{Spiked Wishart model:} \quad y_{ij} = \sqrt{\frac{\lambda}{n}} u_i^* v_j^* + z_{ij}, \quad i \le n,\ j \le m$$

where now the noise is non-symmetric. Some applications of it include:

- *Spiked covariance model:* The signal is $\mathbf{u}^*$ with i.i.d. entries from $P_u = \mathrm{Ber}(\rho)$. The vector $\mathbf{v}^*$ with entries from $P_v = \mathcal{N}(0,1)$ is then interpreted as a noise matrix. The observations $\mathbf{y}_j$, i.e., the columns of $\mathbf{Y}$ are then i.i.d. Gaussian vectors from

$$\mathbf{y}_j \sim \mathcal{N}\Big(0, \boldsymbol{I}_n + \frac{\lambda}{n}\mathbf{u}^* \otimes \mathbf{u}^*\Big).$$

  The goal is to recover the spike $\mathbf{u}^* \otimes \mathbf{u}^*$ of the covariance given these vectors.

- *High-dimensional clustering of m noisy n-dimensional points in k clusters:* Consider the rank-k version of the spiked Wishart model where $\mathbf{u}^* \in \mathbb{R}^{n\times k}$, $\mathbf{v}^* \in \mathbb{R}^{m\times k}$:

$$y_{ij} = \sqrt{\frac{\lambda}{n}} \sum_{\ell=1}^{k} u_{i\ell}^* v_{j\ell}^* + z_{ij}.$$

  Then the columns of $\mathbf{u}^*$ are n-dimensional vectors representing centres of k different clusters. The rows of $\mathbf{v}^*$ are uniform permutations of the k-dimensional vector with a single one $(1,0,\dots,0)$ selecting the cluster to which belong the noisy point $\mathbf{y}_j$ with $j \le m$ (the columns of $\mathbf{Y}$).

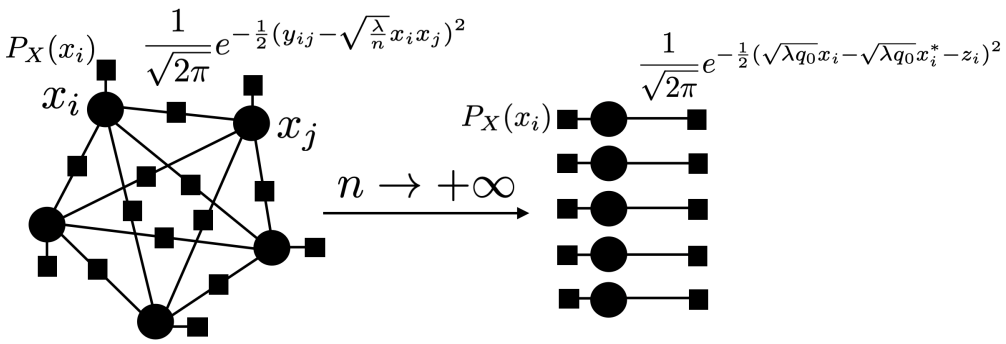


Figure 2: Visualization of the decoupling phenomenon happening in the large size limit: the spiked Wigner model is the original one, the right one is an asymptotically "equivalent" mean-field model with effective Gaussian factors corresponding to independent observations coming from a denoising channel, with SNR given by the original one λ rescaled by the minimizer $q_0(\lambda, \rho)$ of the replica symmetric potential.

3.4 Replica symmetric formula for the mutual information

The main result we will derive is the following asymptotic formula for the mutual information:

Theorem 3 (Replica symmetric formula)**.** *Let the signal prior P_x be bounded with second moment ρ. The mutual information for the spiked Wigner model verifies*

$$\lim_{n\to+\infty} \frac{1}{n} I(\boldsymbol{x}^* \otimes \boldsymbol{x}^*; \boldsymbol{Y}) = \lim_{n\to+\infty} \frac{1}{n} I(\boldsymbol{x}^*; \boldsymbol{Y}) = \inf_{q\in[0,\rho]} i^{(\mathrm{RS})}(q; \lambda, \rho),$$

where, letting $x^* \sim P_x$ and $z \sim \mathcal{N}(0,1)$, the replica symmetric potential *is*

$$i^{(\mathrm{RS})}(q; \lambda, \rho) := \frac{\lambda}{4}(q-\rho)^2 + I(x^*; \sqrt{\lambda q}\, x^* + z) = \frac{\lambda}{4}(q^2+\rho^2) - \mathbb{E}_{x^*,z} \ln \int dP_x(x) e^{\lambda q x x^* + \sqrt{\lambda q} z x - \frac{\lambda q}{2} x^2}.$$

Asymptotic decoupling: Mean-field interpretation

Looking at this formula something peculiar appears: *the computation of the mutual information of the high-dimensional model* (18) *has been reduced to a simple scalar optimization problem.* Surprisingly, in the replica potential appears the mutual information of the denoising problem

$$y_{\lambda q} = \sqrt{\lambda q}\, x^* + z. \tag{22}$$

Lets us look at the stationary condition of the replica symmetric potential. For any finite λ it is not difficult to show that the infimum is attained away from the boundaries, so we compute, based on the I-MMSE formula:

$$\frac{d i^{(\mathrm{RS})}(q)}{dq} = \frac{\lambda}{2}(q-\rho) + \frac{\lambda}{2}\mathrm{MMSE}(x^* | \sqrt{\lambda q}\, x^* + z).$$

Cancelling this derivative we obtain the stationary condition:

$$q = \rho - \mathrm{MMSE}(x^* | \sqrt{\lambda q}\, x^* + z) = \mathbb{E}[x^* \langle x \rangle_{\lambda q}]. \tag{23}$$

We used the consequence (8) of the Nishimori identity. The bracket $\langle \cdot \rangle_{\lambda q}$ is the expectation w.r.t. posterior measure for the denoising problem (22):

$$P(x | y_{\lambda q}(x^*, z)) = \frac{P_x(x) e^{\lambda q x x^* + \sqrt{\lambda q} z x - \frac{\lambda q}{2} x^2}}{\int dP_x(x) e^{\lambda q x x^* + \sqrt{\lambda q} z x - \frac{\lambda q}{2} x^2}}.$$

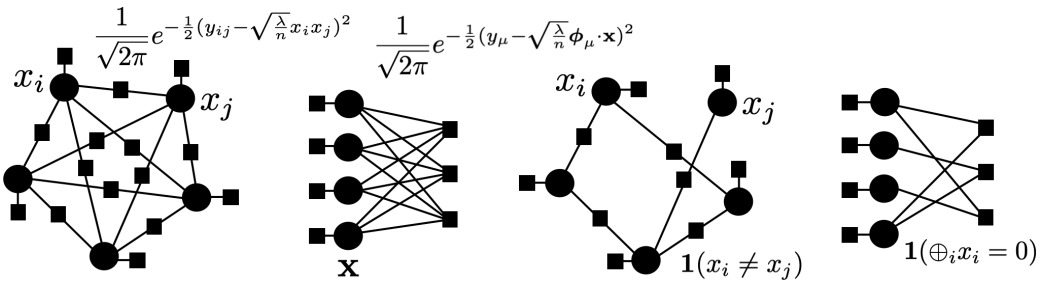


Figure 3: Graphical models/factor graphs associated with different inference and combinatorial optimization problems of a mean-field nature (the two first are dense models, the two last are sparse/tree-like). Starting from left: the factor graph of the spiked Wigner model (the factors connected to a single node represent the independent prior contribution $P_x(x_i)$ encoding, e.g., the domain of the variable etc), of high-dimensional linear regression, of the coloring problem, of a low-density parity-check code.

At least one solution of the self-consistency equation $q = \mathbb{E}[x^* \langle x \rangle_{\lambda q}]$ always exists: $q = 0$. If there are more, the one minimizing the replica symmetric potential, denoted $q_0 = q_0(\lambda, \rho)$, plays a special role: it is linked to the MMSE of the spiked Wigner model as we will see next. So we see that the best error reachable for the original high-dimensional problem reduces in the large system limit to the analysis of a scalar denoising problem with a SNR characterized self-consistently. This type of dimensionality reduction is at the core of mean-field analyses in statistical physics and more generally of modern high-dimensional statistics. Somehow, the high-dimensional limit became a theoretical blessing rather a curse!

The relation between the original high-dimensional problem (the spiked Wigner model) and the denoising one (22) is that following: it can be shown that in the large system limit, any finite subset of components of $\mathbf{x} \sim P(\boldsymbol{x}|\boldsymbol{Y})$ decouple, and their marginals correspond to posterior distributions associated with effective scalar denoising problems of the form (22), each with an independent noise. Roughly this asymptotic decoupling entails

$$\mathbb{E}_{\mathbf{Z}} P(x_i, x_j, \ldots, x_\ell | \boldsymbol{Y}(\mathbf{x}^*, \mathbf{Z})) \approx \mathbb{E}_{z_i} P(x_i | y_{\lambda q}(x_i^*, z_i)) \, \mathbb{E}_{z_j} P(x_j | y_{\lambda q}(x_j^*, z_j)) \ldots \mathbb{E}_{z_\ell} P(x_\ell | y_{\lambda q}(x_\ell^*, z_\ell)) \tag{24}$$

where $P(x_i, x_j, \ldots, x_\ell | \boldsymbol{Y})$ is the marginal posterior distribution for any subset of variables $\{x_i, x_j, \ldots, x_\ell\}$ of cardinal independent of n, obtained from the full joint distribution $P(\mathbf{x}|\boldsymbol{Y})$ by integrating w.r.t. all remaining variables. All noise variables z are independent $\mathcal{N}(0,1)$ capturing the randomness of the data/noise as the level of the mean-field level. Symbol $\approx$ means approximate equality in distribution becoming exact as $n \to \infty$[1]. This is illustrated by Fig. 2.

By the way, now the term "mean-field" model becomes clear: asymptotically, (subset of) variables drawn from the posterior distribution decouple and each one experiences an independent "mean-field" (the factors in the right hand side of Fig. 2) related to a scalar denoising problem that effectively captures the effect of all its complex interactions with the other variables. That the effective problem that appears is a *Gaussian* denoising problem is also expected: this mean-field summarizes very many weak interactions with other variables (weak due to the factor $1/\sqrt{n}$ in (18)), so a kind of "central limit theorem" is expected to take place, yielding an effective Gaussian observation/marginal likelihood for each individual variable.

Such exact dramatic reduction in complexity, going from very high to low dimension, is typical of mean-field models: fully connected models, defined by complete graphical models, and sparse tree-like graphical models, see Fig. 3.

[1]There is a subtlety here: the posterior of the original problem contains information about $\pm\mathbf{x}^*$, i.e., only up to a global sign as the data depends on $\mathbf{x}^* \otimes \mathbf{x}^*$. Instead, the r.h.s. of (24) breaks this invariance. However, when treated correctly, one realizes this is not an issue. In essence, from the theoretical perspective, one can always assume to have access to the global sign of $\mathbf{x}^*$ and thus break this symmetry in the original problem without changing the asymptotic value of the mutual information, MMSE, etc. This is because this sign is just one number to know more, which is nothing compared to the $O(n^2)$ already given by the data, so it does not help much anyway for inference.

Minimum mean-square error

As a consequence of the replica symmetric formula for the mutual information we obtain by application of the I-MMSE identity a simple formula for the (matrix/spike) MMSE:

Theorem 4 (Minimum mean-square error)**.** *Under the same assumptions as in Theorem 3 and for all* (λ, ρ) *such that the minimizer of the replica symmetric potential is unique, in which case we denote this unique minimizer*

$$q_0(\lambda, \rho) := \operatorname*{argmin}_{q \in [0,\rho]} i^{(\mathrm{RS})}(q; \lambda, \rho),$$

the spike-MMSE (or matrix-MMSE) verifies:

$$\lim_{n\to+\infty} \frac{1}{n^2}\mathbb{E}\|\mathbf{x}^* \otimes \mathbf{x}^* - \langle \mathbf{x} \otimes \mathbf{x} \rangle\|^2 = \rho^2 - q_0(\lambda, \rho)^2.$$

Here $\langle \mathbf{x} \otimes \mathbf{x} \rangle = \mathbb{E}[\mathbf{x} \otimes \mathbf{x}|\mathbf{y}]$ *is the spike posterior mean (the MMSE estimator).*

Proof. For a sequence of concave functions with a pointwise limit, the limit of the derivative is the derivative of the limit. Let $I \subset \mathbb{R}$ and $(g_n)_{n\geq 0}$ be a sequence of concave functions on I that converge pointwise to g. Then for any $x \in I$ where $g'(x)$ exists we have $\lim_{n\to+\infty} g_n'(x) = g'(x)$. Indeed by concavity

$$\lim_{n\to+\infty} g_n'(x) \geq \lim_{n\to+\infty} \frac{g_n(x+\varepsilon) - g_n(x)}{\varepsilon} = \frac{g(x+\varepsilon) - g(x)}{\varepsilon} \xrightarrow[\varepsilon\to 0_+]{} g'(x_+).$$

Similarly we obtain $\lim_{n\to+\infty} g_n'(x) \leq g'(x_-)$. If g is differentiable at x then $g'(x_+) = g'(x_-) = g'(x)$, thus the result.

The sequence of functions we have in mind is the mutual information density $\frac{1}{n}I(\mathbf{x}^*; \mathbf{Y})$, which are concave in λ. Indeed by the I-MMSE formula

$$\frac{d}{d\lambda}\frac{1}{n}I(\mathbf{x}^*; \mathbf{Y}) = \frac{1}{2n^2}\sum_{i<j}\mathbb{E}\big[(x_i^* x_j^* - \langle x_i x_j \rangle)^2\big] = \frac{1}{4n^2}\mathbb{E}\|\mathbf{x}^* \otimes \mathbf{x}^* - \langle \mathbf{x} \otimes \mathbf{x} \rangle\|_{\mathrm{F}}^2 + O(1/n). \tag{25}$$

The MMSE is clearly a non-increasing function of the signal-to-noise ratio λ ("information can't hurt"), which can also be verified by direct computation. Consider (λ, ρ) so that the minimizer of the potential is unique: $q_0 = q_0(\lambda, \rho)$. By Theorem 3 the mutual information converges to $i^{(\mathrm{RS})}(q_0; \lambda, \rho)$. It is not difficult to show by direct computation that $q_0(\lambda, \rho) \in (0, \rho)$. Therefore it must be that the q-derivative of $i^{(\mathrm{RS})}$ cancels at $q_0 = q_0(\lambda, \rho)$ and thus, using the I-MMSE relation,

$$\begin{aligned}
\lim_{n\to+\infty} \frac{d}{d\lambda}\frac{1}{n}I(\mathbf{x}^*; \mathbf{Y}) &= \frac{d}{d\lambda} i^{(\mathrm{RS})}(q_0; \lambda, \rho) = \frac{\partial}{\partial\lambda} i^{(\mathrm{RS})}(q_0; \lambda, \rho) \\
&= \frac{(\rho - q_0)^2}{4} + \frac{q_0}{2}\mathrm{MMSE}(x^* | \sqrt{\lambda q_0}x^* + Z) \\
&= \frac{(\rho - q_0)^2}{4} + \frac{q_0}{2}(\rho - q_0) = \frac{\rho^2 - q_0^2}{4}
\end{aligned}$$

using that $q_0(\lambda, \rho)$ verifies the stationary condition (23). Comparing this result with (25) in the limit $n \to +\infty$ ends the proof. □

3.5 Phase diagram: Bayesian-optimal performance and algorithmic hardness

Model (18) may look rather special at first sight. However, a long line of work in the statistical physics of high-dimensional inference has proven that the analysis of prototypical models allow us to understand general mechanisms, similarly to how dissecting the Ising model in physics has paved the way to understand phase transitions and critical phenomena way beyond it. The connection between the Ising model and matrix denoising is actually more than an analogy. Indeed, the spiked Wigner model with signal entries taking binary values $x_i = \pm 1$ is precisely the planted version of the celebrated Sherrington-Kirkpatrick mean-field

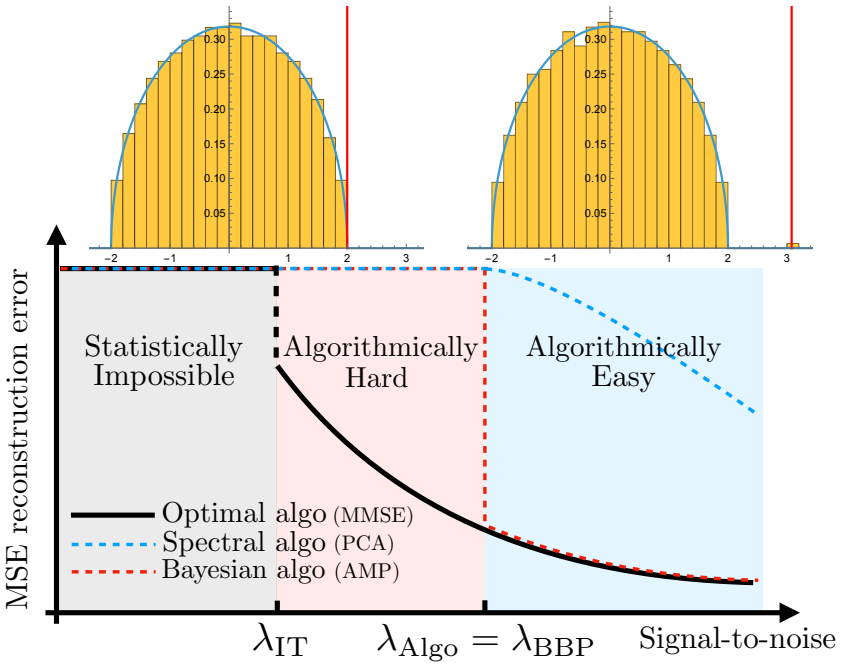


Figure 4: The top illustrates the Ben Arous-Baik-Péché (BBP) transition occurring in the rank-one spiked matrix model. In orange is the histogram of eigenvalues of $\mathbf{Y}$. Before the transition it closely matches Wigner's semicircular law (blue) and its top eigenvalue (red line) sticks to the endpoint of the bulk of eigenvalues. All eigenvectors of $\mathbf{Y}$ have a vanishing $o_n(1)$ overlap with the hidden signal $\mathbf{x}^*$. The BBP transition is marked by an outlier eigenvalue detaching from the bulk. The associated eigenvector aligns non-trivially with $\mathbf{x}^*$ and can thus serve as spectral estimator. At the bottom is the phase diagram of the model considered from the Bayesian (information theoretic) perspective, with prior P_x given by (21). An impossible, hard, and easy inference phases appear for typical realizations of (18) as the signal strength increases, delimited by information theoretic and algorithmic transitions. The behavior of the Bayes-optimal, spectral, and Bayesian approximate message-passing algorithm are shown. For this example of prior P_x, the transitions for the message-passing and spectral algorithms match, but the Bayesian algorithm outperforms the spectral one as it exploits the prior, while the latter does not.

spin glass [**sherrington1975solvable**] (see [**nishimori2001statistical**, **zdeborova2016statistical**] for the mapping), whose analysis is at the core of a fruitful forty-year research field involving physicists and mathematicians [**mezard1990spin**, **Panchenko2013**].

The special features of this spin model inherent to the presence of a planted signal (or equivalently, "on the Nishimori line"), first analysed using spin glass techniques by Nishimori [**nishimori2001statistical**], have triggered numerous works by various communities. Spiked matrix models (i.e., low-rank perturbations of much larger rank matrices, in the present case the noise) are naturally a whole subject in random matrix theory. It was first understood in this context that the presence of the signal may yield a phase transition phenomenon now known as the Ben Arous-Baik-Péché (BBP) transition [**baik2005phase**, **baik2006eigenvalues**, **BENAYCHGEORGES2011**] at a critical value of the signal-to-noise ratio (SNR) λ. This is one of the most studied examples of phase transition in information processing. The top of the right part of Fig. 4 illustrates the BBP transition: for SNR greater than a critical value λ_{BBP}, an outlier eigenvalue of the data $\mathbf{Y}$ escapes the semicircular bulk of eigenvalues associated mostly with the noise component. A simple spectral estimator corresponding to the associated eigenvector then has a non-trivial overlap with $\mathbf{x}^*$. Before this point, however, no eigenvector of the data aligns with the signal, and spectral estimation is thus doomed. In the statistics literature, the spiked Wigner model was introduced as a statistical model for sparse principal components analysis [**johnstone2001distribution**, **johnstone2004sparse**, **zou2006sparse**, **johnstone2012consistency**].

Concerning the Bayesian viewpoint on the model, multiple approaches are now available for the computation of the main information theoretic quantities, in particular the mutual information between signal and data and the minimum achievable mean square error. We mention the replica method [**mezard1990spin**, **lesieur2015mmse**, **lesieur2017constrained**] among the non-rigorous ones, and proofs based on the analysis of message-passing algorithms [**6875223**, **deshpande2017asymptotic**], spatial coupling [**XXT**, **XX˙long**], the interpolation method and its adaptive version [**koradamacris**, **BarbierM17a**, **Barbier˙2019**, **reeves2020information**], the cavity method [**ASS**, **lelarge2019fundamental**], large deviation arguments [**el2018estimation**] or partial differential equations approaches [**mourrat˙Hamilton˙spike**, **chen2022statistical**]. This influential series of works has led to a rich and general picture summarized by Fig. 4.

The phase diagram shown in the figure possesses three distinct regions. In the "impossible phase" at low λ, for a typical large realization of the probabilistic model (18), the data contains a vanishing information about the signal and thus inference, whether computationally efficient or not, is impossible. Consequently, even the Bayes-optimal estimator (corresponding to the Bayes posterior mean, black solid curve) yields trivially bad reconstruction error. Beyond the information theoretic transition $\lambda_{\rm IT}$ the picture changes: here, the data typically contain information and inference becomes in theory possible. However, no practical algorithm running with a sub-exponential complexity in N is known. Those we have at hand, including the aforementioned spectral estimator and a Bayesian, prior-aware, approximate message-passing (AMP) algorithm [**kabashima2003cdma**, **donoho2009message**, **rangan2012iterative**, **lesieur2017constrained**] are all failing. Finally, beyond the algorithmic threshold $\lambda_{\rm Algo}$ (which, for the choice of prior in Fig. 4, matches $\lambda_{\rm BBP}$), the signal is strong enough to be efficiently extracted from the data and, in particular, the Bayesian AMP algorithm suddenly becomes optimal in spite of its low computational cost, following a phase transition in its performance. In this region inference is thus "easy"

Such a phase diagram with a hard region (also called computational-to-statistical gap) and associated discontinuous transitions in algorithms performance, is not specific to matrix denoising but is actually typical in high-dimensional inference [**zdeborova2016statistical**, **BarbierGLM-PNAS**]. Their study is the central question in average-case complexity theory in modern computer science. And web-of-reductions exist that prove that if the hard phase is fundamental (in the sense that there exists no polynomial-in-N complexity algorithm for signal extraction in this phase), then a plethora of other interconnected inference problems must also be hard [**brennan2018reducibility**, **brennan2020reducibility**]. The analyses of prototypical matrix denoising models therefore have a conceptual impact that goes way beyond their specific instance.

4 (Lect. 4, 5 and 6) Random matrix theory 101: Performance of spectral PCA

For this part I recommend the books by Livan et al. and the one by Potters and Bouchaud.

In the last lectures we have quantified what is the *optimal* performance one can aim for if using the correct prior knowledge about the signal (we will derive this using the replica method from spin statistical mechanics later on). The issue that is that the minimum mean-squared estimator is the posterior mean, whose computation is in general intractable in high dimension (we will see in the last lectures that it becomes so in some regimes of SNR). A natural question is then:

Can we provide a simpler inference procedure than computing the posterior mean, which will thus be sub-optimal, but that is at least computationally efficient?

The answer is yes, this is *spectral PCA*, which simply corresponds to taking the leading eigenvector $\mathbf{x}_{\rm PCA} = \mathbf{v}_1(\mathbf{Y})$ of the data matrix as estimator, see Fig. 4. This has a complexity of order $O(N^3)$, the number of operations needed to diagonalise the data, which is therefore much less than the $O(e^N)$ operations required in general to compute the posterior mean (as one needs to sum over all possible signal configurations). To study the performance os spectral PCA we need to use Random Matrix Theory (RMT).

RMT studies the statistical properties of matrices with random entries. It has found applications across physics, statistics, and data science. It was born in physics. Eugene Wigner studied energy levels of heavy atomic nuclei. He modelled complex quantum systems by random Hamiltonians (large Hermitian matrices).

Key insight: Statistical properties of eigenvalues follow universal laws.

Surprise: Simple models of random matrices accurately predict various *statistics* of energy levels of complex quantum systems: a strong universality seems to hold.

4.1 The Gaussian orthogonal ensemble and the semicircle law

The Gaussian orthogonal ensemble (GOE, or Wigner ensemble) of random matrices is the simplest and most studind ensemble of random real symmetric matrices (we will always consider real matrices, which are symmetric so that their eigenvalues are real).

Definition 4 (GOE)**.** *A random $n \times n$ symmetrix matrix $\mathbf{H} = \mathbf{H}^\intercal$ belongs to the GOE if its upper-triangular elements are independent:*

$$(\mathbf{H})_{ij} = h_{ij} = h_{ji} \sim \begin{cases} \mathcal{N}(0, 2/n) & i = j \\ \mathcal{N}(0, 1/n) & i < j \end{cases}$$

Equivalently, its joint density reads

$$P(\mathbf{H}) \propto \exp\Big(-\frac{n}{4} \operatorname{Tr}(\mathbf{H}^2) \Big)$$

GOE matrices are *rotationally invariant*, meaning that any rotation of a GOE matrix has the same probability to occur as the non-rotated one: for any $n \times n$ orthogonal matrix $\mathbf{O}$,

$$P(\mathbf{OHO}^\intercal) \propto \exp\Big(-\frac{n}{4} \operatorname{Tr}(\mathbf{OHO}^\intercal \mathbf{OHO}^\intercal) \Big) = \exp\Big(-\frac{n}{4} \operatorname{Tr}(\mathbf{H}^2) \Big)$$

by cyclicity of the trace. In other words, $\mathbf{OHO}^\intercal$ has the same law as $\mathbf{H}$. This ensemble of random matrices is the only rotationally-invariant one, whose entries are independent (up to symmetry). One of the most basic object of interest in RMT is the *spectral density*, i.e., the limit of the empirical distribution of eigenvalues.

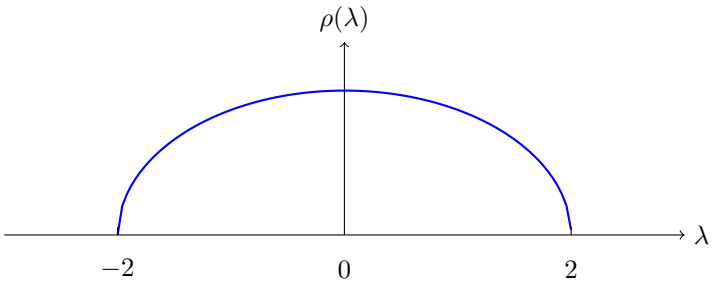
For large n, the empirical eigenvalue distribution

$$\rho_{\mathbf{H}}(\lambda) = \frac{1}{n} \sum_{i \le n} \delta(\lambda_i(\mathbf{H}) - \lambda)$$

of a random matrix $\mathbf{H}$ drawn from the GOE ensemble converges with high probability as $n \to \infty$ to the so-called semicircle law.

Theorem 5 (Wigner's Semicircle Law)**.** *The eigenvalue empirical density of a GOE matrix a.s. converges as $n \to \infty$ to the semicircle law*

$$\lim_{n \to \infty} \rho_{\mathbf{H}}(\lambda) = \rho_{\mathrm{sc}}(\lambda) = \begin{cases} \frac{1}{2\pi} \sqrt{4 - \lambda^2} & |\lambda| \le 2 \\ 0 & |\lambda| > 2 \end{cases} \qquad a.s.$$



4.2 The Wishart ensemble and Marcenko-Pastur law

The Wishart ensemble is fundamental in multivariate statistics, describing the distribution of sample covariance matrices associated with "noise" (i.e., Gaussian matrices, or more generally, with i.i.d. entries by "universality").

Definition 5 (Wishart matrix)**.** *Let* $\mathbf{X}$ *be an* $n \times p$ *random matrix with i.i.d. entries* $X_{ij} \sim \mathcal{N}(0,1)$*, where* n *is the number of samples,* p *is the number of features for each sample. The* $p \times p$ *sample-covariance matrix*

$$\mathbf{W} = \frac{1}{n}\mathbf{X}^\top \mathbf{X}$$

is called a ***Wishart matrix*** *(or belongs to the Wishart ensemble).*

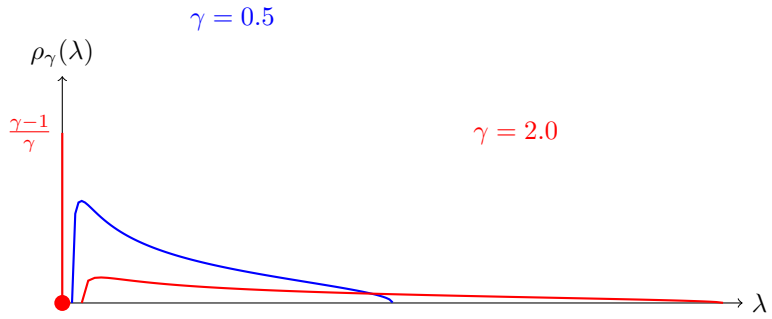
- $\mathbf{W}$ is symmetric and positive semi-definite
- Eigenvalues $\lambda_i(\mathbf{W}) \geq 0$ represent variances along principal axes (eigenvectors)
- When p is fixed and $n \to \infty$, $\mathbf{W} \to \mathbf{I}_p$ (law of large numbers)
- Interesting high-dimensional regime: $n, p \to \infty$ with $p/n \to \gamma > 0$

The eigenvalue distribution converges to the Marcenko-Pastur law.

Theorem 6 (Marcenko-Pastur Law)**.** *As* $n, p \to \infty$ *with* $p/n \to \gamma \in (0, \infty)$*, the eigenvalue density of* $\mathbf{W}$ *a.s. converges to the Marcenko-Pastur distribution:*

$$\lim_{n,p\to\infty} \rho_{\mathbf{W}} = \rho_\gamma(\lambda) = \begin{cases} \frac{1}{2\pi\gamma\lambda}\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)} & \lambda \in [\lambda_-, \lambda_+] \\ 0 & \textit{otherwise} \end{cases} + (1 - \gamma^{-1})1(\gamma > 1)\,\delta(\lambda), \quad a.s.$$

where $\lambda_\pm = (1 \pm \sqrt{\gamma})^2$*.*



PCA significance and RMT denoising: Distinguish true signals from noise eigenvalues. Eigenvalues below λ_+ are related to noise directions/eigenvectors, see more details later.

4.3 Resolvent and Stieltjes transform

Definition 6 (Resolvent)**.** *The resolvent of an* $n \times n$ *matrix* $\mathbf{H}$ *is:*

$$\mathbf{G}_\mathbf{H}(z) = (z\mathbf{I} - \mathbf{H})^{-1}, \quad z \in \mathbb{C} \setminus \{\lambda_i(\mathbf{H})\}_{i \leq n}$$

Definition 7 (Stieltjes transform)**.** *The Stieltjes transform of the spectral measure* ρ_H *is:*

$$g_H(z) = \int_{\mathbb{R}} \frac{\rho_H(\lambda)}{z - \lambda} d\lambda = \lim_{n\to\infty} \mathbb{E}_\mathbf{H} \frac{1}{n} \operatorname{Tr} \mathbf{G}_\mathbf{H}(z), \quad z \in \mathbb{C} \setminus \operatorname{Supp}(\rho)$$

The importance of the Stieljes transform resides in the fact that it is usually much simpler to compute than directly getting ρ_H for a certain random matrix ensemble (such as the GOE), but contains the same information. Indeed, it is the moment generating function of the spectral density:

$$\frac{\partial g_H(z)}{\partial (1/z)^{k+1}}\Big|_{z=\infty} = (k+1)! \int \lambda^k \rho_H(\lambda) d\lambda. \tag{26}$$

Importantly, it allows to access it via the following relation.

Theorem 7 (Stieltjes inversion formula)**.** *The spectral density can be recovered from the Stieltjes transform: letting $\Im(\cdot)$ be the imaginary part,*

$$\rho_H(\lambda) = \lim_{\epsilon \to 0^+} \frac{1}{\pi} \Im g_H(\lambda - i\epsilon)$$

Proof. We drop the subscript H. Consider the imaginary part of the Stieltjes transform:

$$\Im g(\lambda - i\epsilon) = \Im \int_{\mathbb{R}} \frac{\rho(\mu)}{(\lambda - \mu) - i\epsilon} d\mu = \int_{\mathbb{R}} \rho(\mu) \frac{\epsilon}{(\lambda - \mu)^2 + \epsilon^2} d\mu$$

The kernel $\epsilon/((\lambda - \mu)^2 + \epsilon^2)$ is an approximation to the delta function (Poisson kernel). As $\epsilon \to 0^+$, it converges to $\pi\delta(\lambda - \mu)$, hence:

$$\lim_{\epsilon \to 0^+} \frac{1}{\pi} \Im g(\lambda - i\epsilon) = \rho(\lambda)$$

□

The Stieltjes transforms of the semicirle law and Marcenko-Pastur respectively read

$$g_{\text{sc}}(z) = \frac{z \pm \sqrt{z^2 - 4}}{2}, \qquad g_\gamma(z) = \frac{z - 1 + \gamma \pm \sqrt{(z - 1 + \gamma)^2 - 4\gamma z}}{2\gamma z}.$$

The spectral density is recovered via Stieltjes inversion.

Recovering the semicircle law from the Stieljes transform

Set $z = \lambda - i\epsilon$. We have

$$2g(\lambda - i\epsilon) = \lambda - i\epsilon \pm \sqrt{\lambda^2 - \epsilon^2 - 4 - i2\lambda\epsilon}. \tag{27}$$

Letting $\tau = \lambda^2 - \epsilon^2 - 4 - i2\lambda\epsilon$ and using that

$$\sqrt{\tau} = \frac{1}{\sqrt{2}} \sqrt{|\tau| + \Re\tau} + i \frac{\text{sign}(\Im\tau)}{\sqrt{2}} \sqrt{|\tau| - \Re\tau}$$

with $|\tau| = \sqrt{(\Re\tau)^2 + (\Im\tau)^2}$, we get

$$2\Im g(\lambda - i\epsilon) = -\epsilon \pm \frac{\text{sign}(\Im\tau)}{\sqrt{2}} \sqrt{|\tau| - \Re\tau} \to \pm \frac{\text{sign}(\Im\tau)}{\sqrt{2}} \sqrt{|\tau| - \Re\tau}$$

Then, if $|\tau| = \Re\tau$ when $\epsilon = 0$, i.e., $|\lambda| \geq 2$, then the above is 0 and thus the density too. If $|\lambda| < 2$ we instead get

$$2\Im g(\lambda - i\epsilon) \to -\frac{\text{sign}(\Im\tau)}{\sqrt{2}} \sqrt{|\lambda^2 - 4| - \lambda^2 + 4} = \pm \text{sign}(-\lambda) \sqrt{4 - \lambda^2}.$$

Because this is a probability density it must stay non-negative, which allows to select the correct sign depending on whether λ is positive or negative. Dividing by π we finally reach via the Stieltjes inversion formula:

$$\rho(\lambda) = \lim_{\epsilon \to 0^+} \frac{1}{\pi} \Im g(\lambda) = \frac{1}{2\pi} \sqrt{4 - \lambda^2} \quad \text{for} \quad |\lambda| < 2, \quad 0 \quad \text{else.}$$

4.4 Derivation of the Stieljes transform for the GOE via the cavity method

Now that we know how to get the spectral desity from the Stieljes transform, the question is how to obtian the latter? We can derive it easily for GOE or Wishart matrices using the cavity method and the Schur complement formula. The Schur complement, also called inversion by partitioning, relates the blocks of the inverse of a matrix to the inverse of blocks of the original matrix.

Theorem 8 (Schur complement formula)**.** *Let* $\mathbf{M}$ *be an invertible matrix which we divide in four blocks as*

$$\mathbf{M} = \begin{pmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{pmatrix} \quad \textit{and} \quad \mathbf{M}^{-1} = \mathbf{Q} = \begin{pmatrix} \mathbf{Q}_{11} & \mathbf{Q}_{12} \\ \mathbf{Q}_{21} & \mathbf{Q}_{22} \end{pmatrix},$$

where $\mathbf{M}_{11}$ *is* $n \times n$*,* $\mathbf{M}_{12}$ *is* $n \times (n-t)$*,* $\mathbf{M}_{21}$ *is* $(n-t) \times n$*,* $\mathbf{M}_{22}$ *is* $(n-t) \times (n-t)$*, and* $\mathbf{M}_{22}$ *is invertible. The integer* t *can take any values from 1 to* $n-1$*. Then the upper left* $n \times n$ *block of* Q *is given by*

$$\mathbf{Q}_{11}^{-1} = \mathbf{M}_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21},$$

where the right hand side is called the Schur complement of the block $\mathbf{M}_{22}$ *of the matrix* $\mathbf{M}$*.*

Now, consider an $n \times n$ GOE matrix $\mathbf{H}_n$. Let

$$\mathbf{M}_n = z\mathbf{I}_n - \mathbf{H}_n, \quad \text{so} \quad \mathbf{M}^{-1} = \mathbf{G}_{\mathbf{H}_n}(z) = \mathbf{G}_n(z).$$

The cavity method is about comparing a system of size n and $n-1$ and exploit that their statistics converge to the same when the system size $n \to \infty$ in order to obtain self-consistent equations of these statistics of interest. Partition $\mathbf{M}$ as

$$\mathbf{M}_n = \begin{pmatrix} z - h_{11} & \mathbf{v} \\ \mathbf{v}^\intercal & \mathbf{M}_{n-1} \end{pmatrix}$$

where

$$\mathbf{M}_{n-1} = \Big(z\sqrt{\frac{n}{n-1}}\mathbf{I}_{n-1} - \mathbf{H}_{n-1}\Big)\sqrt{\frac{n-1}{n}}$$

is $(n-1) \times (n-1)$, $\mathbf{v}$ is $(n-1)$-dimensional vector with $v_i \sim \mathcal{N}(0, 1/n)$, and $h_{nn} \sim \mathcal{N}(0, 2/n)$. By the Schur complement formula,

$$\frac{1}{(\mathbf{G}_n(z))_{11}} = z - h_{11} - \sqrt{\frac{n}{n-1}}\mathbf{v}^\intercal \mathbf{G}_{n-1}\Big(z\sqrt{\frac{n}{n-1}}\Big)\mathbf{v} \quad \text{where} \quad \mathbf{G}_{n-1}(z) = (z\mathbf{I}_{n-1} - \mathbf{H}_{n-1})^{-1}.$$

Self-consistent equation exploiting concentration in the large size limit

As $n \to \infty$:

1. $\mathbf{v}^\intercal \mathbf{G}_{n-1}(z)\mathbf{v} \approx \frac{1}{n}\mathrm{Tr}\mathbf{G}_n \approx g(z)$ by independence
2. $h_{11} \sim \mathcal{N}(0, 2/n)$ is negligible
3. By the equation above and two last points, we can see that as $n \to \infty$, $(\mathbf{G}_n)_{11}$ concentrates and is equal to $(\mathbf{G}_n)_{ii}$ by rotational invariance, and thus also to $\frac{1}{n}\mathrm{Tr}\mathbf{G}_n \approx g(z)$

Then we obtain the self-consistent equation

$$g(z)^{-1} = z - g(z) \quad \Rightarrow \quad 1 - zg(z) + g(z)^2 = 0 \quad \Rightarrow \quad g(z) = g_{\mathrm{sc}}(z).$$

4.5 BBP transition: Detecting the presence of information in the data

We aim at understanding how a rank-one matrix (modelling a structured hidden information) of a full-rank noise matrix perturbs it. In particular, can this perturbation be reconstructed from the sum of the two (i.e., the observed data)?

Problem: Distinguish noise from signal (if any).

Why RMT is useful: RMT predicts the properties (such as the eigenvalue distribution) for pure noise random matrices, which can then serve as null hypothesis to be compared with in order to detect whether some additional information is contained in the data, apart from the noise component.

Rescaling the spiked Wigner model

From the RMT perspective, the spiked Wigner model simply models a rank-one perturbation of a GOE matrix. It will be convenient to rescale the model with a global $1\sqrt{n}$, i.e.,

$$\mathbf{D} = \tau\, \mathbf{u}\mathbf{u}^\intercal + \mathbf{H}$$

where, as before,

- $\mathbf{D} = \mathbf{Y}/\sqrt{n} \in \mathbb{R}^{n\times n}$ is the rescaled data matrix
- $\mathbf{H} = \mathbf{Z}/\sqrt{n}$ is standard GOE noise matrix
- $\mathbf{u} = \mathbf{x}^*/\sqrt{n} \in \mathbb{R}^n$ is now a deterministic unit vector ($\|\mathbf{u}\| = 1$) representing the low-rank information to extract (which is a bit more general than before)
- $\tau = \sqrt{\lambda} > 0$ controls again the signal strength

The reason we changed notations is that for the statistical mechanics analysis, it is convenient to work with $O(1)$ signal and noise variables (which implies matrices with $O(\sqrt{N})$ spectral norm), while for the RMT analysis it is convenient to work with matrices whose spectral norm is $O(1)$ (i.e., with a spectrum that remains with bounded support in the large size limit $n \to \infty$).

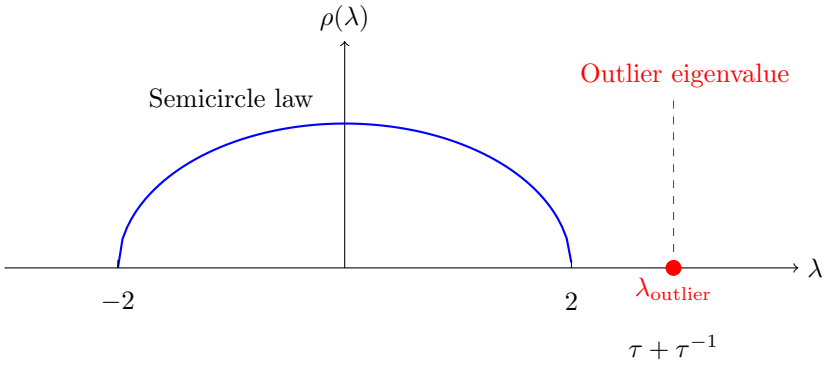
In RMT, this is the most basic signal plus noise problem, which is solved by the following result. The same problem can be solved similarly for a much broader class of random noise ensembles, including Wishart matrices (in this case, the model is more naturally connected to covariance denoising and thus PCA).

Theorem 9 (Ben Arous-Baik-Péché (BBP) transition)**.** *The largest eigenvalue $\lambda_1 = \lambda_1(\mathbf{D})$ of $\mathbf{D}$ satisfies:*

- *If $\tau \le 1 =: \tau_{\text{BBP}}$, $\lambda_1 \to 2$ almost surely as $n \to \infty$*
- *If $\tau > 1$, $\lambda_1 \to \tau + \tau^{-1} > 2$ almost surely*

Moreover, for $\tau > \tau_{\text{BBP}} = 1$, the eigenvector $\mathbf{v}_1$ of $\mathbf{D}$ associated with λ_1 almost surely has a non-trivial overlap with the perturbation $\mathbf{u}$:

$$\lim_{n\to\infty} |\langle \mathbf{v}_1, \mathbf{u}\rangle|^2 = 1 - \tau^{-2} \quad a.s.$$



This result on the overlap gives the performance of *spectral PCA*, i.e., the estimator $\hat{\mathbf{u}}_{\text{PCA}} = \mathbf{v}_1$ given by the leading eigenvector of the data. Notice that it is *not* a Bayesian procedure: it does not exploit any possible prior information about $\mathbf{u}$, only that its norm is 1.

NOT REALLY, MENTION THE RESCALING OF NORM : From the Bayes-optimal performace we derive in previous lectures, one can check by comparison with the MMSE formula that *spectral PCA is optimal (yields maximum overlap, or equivalently minimum MSE) when the prior is Gaussian.* Indeed, a Gaussian prior is highly uninformative and therefore does not bring more information than the data (aprt on the norm of the signal, which is also exploited by spectral PCA)

4.6 Resolvent analysis: Outlier location and critical SNR

Let us (kind of) rigorously show the BBP results, starting with the outlier location and identifying when the transition in the SNR occurs (when does the outlier escape the boundary of the semicircle due to noise).

Theorem 10 (Sherman-Morrison formula)**.** *Given a rank-one matrix* $\mathbf{xy}^\intercal$ *of same dimension as the invertible matrix* $\mathbf{A}$*,*

$$(\mathbf{A} + \mathbf{xy}^\intercal)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{xy}^\intercal\mathbf{A}^{-1}}{1 + \mathbf{y}^\intercal\mathbf{A}^{-1}\mathbf{x}}$$

The resolvent of $\mathbf{D}$ is:

$$\mathbf{G_D}(z) = (z\mathbf{I} - \mathbf{H} - \tau\mathbf{u}\mathbf{u}^\intercal)^{-1}$$

Using Sherman-Morrison:

$$\mathbf{G_D}(z) = \mathbf{G_H}(z) + \frac{\tau\mathbf{G_H}(z)\mathbf{u}\mathbf{u}^\intercal\mathbf{G_H}(z)}{1 - \tau\mathbf{u}^\intercal\mathbf{G_H}(z)\mathbf{u}}$$

We are looking for an outlier eigenvalue $\lambda > 2 = \lim_{n\to\infty}\lambda_1(\mathbf{H})$. For $z > 2$, $\mathbf{G_D}(z)$ is regular. The outlier eigenvalue $\lambda_1 = \lambda_1(\mathbf{D})$ thus corresponds to a pole where

$$1 - \tau\mathbf{u}^\intercal\mathbf{G_H}(\lambda_1)\mathbf{u} = 0 \tag{28}$$

In the large n limit, for λ_1 outside $[-2, 2]$, we have by independence of $\mathbf{u}$ and $\mathbf{G_H}(z)$ that

$$\mathbf{u}^\intercal\mathbf{G_H}(\lambda_1)\mathbf{u} \to g_{\mathrm{sc}}(\lambda_1) = \frac{z \pm \sqrt{\lambda_1^2 - 4}}{2}$$

The pole equation (28) asymptotically becomes

$$1 - \tau g_{\mathrm{sc}}(\lambda_1) = 0 \implies g_{\mathrm{sc}}(\lambda_1) = \tau^{-1}$$

Solving for λ_1 gives $\lambda_1 = \tau + \tau^{-1}$ if $\tau > 1$.

This solution is valid only when:

- $\lambda_1 > 2$ (outside the semicircle support)
- The square root $\sqrt{\lambda_1^2 - 4}$ is real

These conditions are satisfied iff:

$$\tau + \tau^{-1} > 2 \iff \tau > 1$$

For $\tau \leq 1$, the pole equation has no solution outside $[-2, 2]$, so no outlier appears. The BBP transition where the outlier appears thus happens at $\tau = 1$.

4.7 Eigenvector overlap: Performance of spectral PCA

Starting from the spectral decomposition of the resolvent (which has the same eigenvectors $(\mathbf{v}_i)$ as $\mathbf{D}$),

$$\mathbf{G_D}(z) = \sum_{i\leq n}\frac{\mathbf{v}_i\mathbf{v}_i^\intercal}{z - \lambda_i(\mathbf{D})}$$

we see that for $\tau > 1$, the eigenvector overlap is given by:

$$|\langle\mathbf{v}_1, \mathbf{u}\rangle|^2 = \lim_{z\to\lambda_1}(z - \lambda_1)\mathbf{u}^\intercal\mathbf{G_D}(z)\mathbf{u}$$

because only the desired term diverges in this limit. Using Sherman-Morrison

$$\mathbf{G_D}(z) = \mathbf{G_H}(z) + \frac{\tau\mathbf{G_H}(z)\mathbf{u}\mathbf{u}^\intercal\mathbf{G_H}(z)}{1 - \tau\mathbf{u}^\intercal\mathbf{G_H}(z)\mathbf{u}}$$

we have

$$\lim_{z\to\lambda_1}(z-\lambda_1)\mathbf{u}^\intercal \mathbf{G}_{\mathbf{D}}(z)\mathbf{u} \approx \lim_{z\to\lambda_1}(z-\lambda_1)\Big(g_{\mathrm{sc}}(z)+\tau\frac{g_{\mathrm{sc}}(z)^2}{1-\tau g_{\mathrm{sc}}(z)}\Big).$$

Using l'Hopital rule to take care of the 0/0 limit, we finally get (using $\tau^{-1}=g_{\mathrm{sc}}(\lambda_1)$)

$$\lim_{n\to\infty}|\langle \mathbf{v}_1,\mathbf{u}\rangle|^2 = -\frac{g_{\mathrm{sc}}(\lambda_1)^2}{g'_{\mathrm{sc}}(\lambda_1)} = 1-\tau^{-2}$$

when $\tau>1$, 0 else.

Applications of BBP transition

- **Signal detection**: Detect weak signals in noisy high-dimensional data
- **PCA**: Determine significance of principal components
- **Community detection**: Identify communities in networks
- **Neuroscience**: Detect correlated neuron assemblies
- **Financial markets**: Identify significant factors in stock returns
- BBP phase transition occurs for spiked Wishart models too (and many other spiked models)

5 (Lect. 7 and 8) Statistical mechanics 101: Spin glasses and links with Bayesian inference

Statistical mechanics provides powerful tools for understanding complex systems with many interacting components. One paradigmatic model is the Ising model.

5.1 Fully connected ferromagnetic Ising model: The paradigmatic spin model

- **Spins**: $\sigma=(\sigma_1,\dots,\sigma_n)\in\{-1,1\}^n$ (microstate configuration)
- **Hamiltonian** (energy function):

$$H_n(\sigma) = -\frac{J}{n}\sum_{1\le i<j\le n}\sigma_i\sigma_j - h\sum_{i=1}^n\sigma_i$$

  - J: Coupling strength
  - h: External magnetic field

- **Gibbs-Boltzmann distribution and partition function**:

$$P_n(\sigma)=\frac{1}{Z_n}e^{-\beta H_n(\sigma)},\quad Z_n=\sum_{\sigma\in\{-1,1\}^n}e^{-\beta H_n(\sigma)},\qquad \beta=\frac{1}{T}$$

  where T is temperature and Z_n is the partition function

- **Order parameter**: macroscopic description (tells in which thermodynamic phase the system is):

$$\text{magnetization}\quad m_n(\sigma)=\frac{1}{n}\sum_{i=1}^n\sigma_i \xrightarrow{n\to\infty} m$$

- **Macrostate**: set of microstates with same value of order parameter:

$$\big\{\sigma\in\{-1,1\}^n : m_n(\sigma)\in[\bar m,\bar m+\epsilon]\big\}$$

- **Thermal average**:
$$\langle g(\sigma)\rangle = \sum_{\sigma\in\{-1,1\}^n} P_n(\sigma)g(\sigma)$$

- **Free energy, and the thermodynamic limit**: contains all thermodynamic information
$$F_n = -\frac{1}{\beta n}\ln Z_n, \quad f = \lim_{n\to\infty} F_n$$

- **Thermodynamic identities**: derivatives of free energy with respect to control parameters (here the external field) give access to order parameters and their fluctuations:
$$\frac{\partial}{\partial h} = -\langle m_n\rangle, \qquad \frac{\partial^2 F_n}{\partial h^2} = -\beta n\cdot \mathrm{Var}(m_n)$$

Phase diagram

- **Second-order transition in** β: Continuous change in magnetization

- **Critical temperature and thermodynamic phases** β_c: Separates ordered/ferromagnetic phase, i.e. $m \neq 0$, at $\beta > \beta_c$ and disordered/paramagnetic phase, i.e. $m = 0$ at $\beta < \beta_c$.

  Phase transition = non-analytic point in f as a function of control parameter (here β)

- For $\beta < \beta_c$ and $h = 0$: Spontaneous symmetry breaking

5.2 The Sherrington-Kirkpatrick (SK) model: The paradigmatic spin glass model

- **Quenched disorder (random couplings)**: $J_{ij} \sim \mathcal{N}(J_0/n, 1/n)$ (quenched disorder), with ferromagnetic bias $J_0 > 0$; in the last equality we extracted the ferromagnetic component J_0 from the pure noise disorder $z_{ij} \sim \mathcal{N}(0,1)$.

- **Hamiltonian**:
$$H_n(\sigma) = -\sum_{1\le i<j\le n} J_{ij}\sigma_i\sigma_j = -\sum_{1\le i<j\le n}\sigma_i\sigma_j\Big(\frac{J_0}{n} + \frac{z_{ij}}{\sqrt{n}}\Big) \tag{29}$$

- **Partition function**:
$$Z_n = \sum_{\sigma\in\{-1,1\}^n} e^{-\beta H_n(\sigma)}$$

- **Overlap** between configurations (i.e., independent samples, called *replicas*, from the same Gibbs-Boltzmann measure $\propto e^{-\beta H_n(\cdot)}$, i.e., which share the same quenched disorder J):
$$q_n(\sigma^1,\sigma^2) = \frac{1}{n}\sum_{i=1}^n \sigma_i^1\sigma_i^2$$

- **Average free energy**:
$$F_n = -\frac{1}{\beta n}\mathbb{E}_J \ln Z_n(J).$$

  We see now an additional expectation w.r.t. the quenched disorder J. The free energy is of paramount importance. This is because from this quantity we can then locate the phase transitions: these are points where the large n limit of the free energy is non-analytic. From it we can also derive all relevant macroscopic/global quantities of interest (average energy density, magnetization, etc), by simply taking derivatives with respect to control parameters (β, external magnetic field, etc).

- **Self-averaging / concentration property**:

$$-\frac{1}{\beta n}\ln Z_n(J) = -\frac{1}{\beta n}\mathbb{E}_J \ln Z_n(J) + o_n(1),$$

where $o_n(1) \to 0$ with high probability w.r.t. the disorder J. Studying averages are enough to describe fixed instances of the model (i.e., J). This justifies an *ensemble analysis*: instead of considering a particular realization of the disorder defining the Hamiltonian of the model, we can average over it and this will give the same results. This is the reason why statistical mechanics makes sense.

Phase diagram

- Paramagnetic phase at sufficiently low β and low J_0: $m = q^{12} = 0$
- Ferromagnetic phase at sufficiently high J_0: $m \neq 0, q^{12} = 0$
- Spin glass phase at sufficiently high β and low J_0: $m = 0, q^{12} \neq 0$

5.3 Linking Bayesian inference and statistical mechanics

We suppose that we are in a *Bayesian optimal setting* where the prior P_X as well as the noise distribution are known so that the true posterior is used by the statistician. Studying this ideal setting allows to delimit fundamental performance bounds for any inference algorithm given the same data. Bayes' formula gives

$$P(\boldsymbol{x}|\boldsymbol{y}) \propto \prod_{i=1}^{n} P_X(x_i) \frac{1}{(2\pi)^{\frac{n(n-1)}{4}}} \exp\Big\{-\frac{1}{2}\sum_{1\leq i<j\leq n}\Big(y_{ij} - \sqrt{\frac{\lambda}{n}}x_i x_j\Big)^2\Big\} \tag{30}$$

$$= \frac{1}{\mathcal{Z}_n(\boldsymbol{y})}\prod_{i=1}^{n} P_X(x_i)\exp\{-\mathcal{H}(\boldsymbol{x};\boldsymbol{y})\}, \tag{31}$$

where, after simplifying all the x-independent terms with the normalization, the Hamiltonian and partition function are

$$\mathcal{H}(\boldsymbol{x};\boldsymbol{y}) = \sum_{i<j}\Big(\frac{\lambda}{2n}x_i^2 x_j^2 - y_{ij}\sqrt{\frac{\lambda}{n}}x_i x_j\Big) \tag{32}$$

$$= \sum_{i<j}\Big(\frac{\lambda}{2n}x_i^2 x_j^2 - \frac{\lambda}{n}x_i^* x_j^* x_i x_j - \sqrt{\frac{\lambda}{n}}z_{ij}x_i x_j\Big), \tag{33}$$

$$\mathcal{Z}_n(\boldsymbol{y}) = \int \prod_{i=1}^{n} dP_X(x_i)\exp\{-\mathcal{H}(\boldsymbol{x};\boldsymbol{y})\}.$$

We used the definition of $\mathbf{y} = \mathbf{y}(\mathbf{x}^*, \mathbf{z})$ to express the Hamiltonian as a function of the independent variables. Due to these simplifications, you can easily check that the more convenient free energy expression

$$f_n := -\frac{1}{n}\mathbb{E}\ln \mathcal{Z}_n(\mathbf{D}) = -\frac{1}{n}\mathbb{E}\ln \mathcal{Z}_n(\mathbf{X}^*, \mathbf{Z}),$$

is related to the Shannon entropy of the data and the mutual information through

$$\frac{1}{n}H(\mathbf{Y}) = f_n + \frac{n-1}{4}\ln(2\pi e) + \frac{\rho^2\lambda}{4}\frac{n-1}{n}, \qquad \frac{1}{n}I(\mathbf{X}^*;\mathbf{Y}) = f_n + \frac{\rho^2\lambda}{4}\frac{n-1}{n}. \tag{34}$$

The spiked Wigner model is a "planted" SK model

From the above mappings, it is clear that high-dimensional Bayesian inference is naturally cast as a statistical mechanics model. This is more than an analogy. e.g., the spiked Wigner model *is* a a version of the spin glass SK model: this is seen by comparing (33) with (29) through the identification of the planted (signal) term $J_0 \leftrightarrow \lambda x_i^* x_j^*$ and the noise term, and noting that $x_i^2 x_j^2 = 1$ (so can be absorbed in the normalisation) if the variables are binary as in the SK model. Note that the scaling of each terms are the same in both Hamiltonians.

5.4 Dictionary: Statistical mechanics ↔ inference

Statistical mechanics	Bayesian inference
Spins $\sigma_i \in \{-1, 1\}$ / degrees of freedom / particles positions	Variables / bits / unknowns $X_i \in \mathbb{R}$
Quenched disorder J_{ij}	Data / observations y_{ij}
Hamiltonian / energy $H_{\boldsymbol{J}}(\boldsymbol{\sigma})$	Negative log-posterior $-\log P(\mathbf{x}\|\mathbf{y})$
Gibbs-Boltzmann distribution $\frac{1}{Z(\boldsymbol{J})}e^{-\beta H_J(\boldsymbol{\sigma})}$	Posterior distribution $P(\mathbf{x}\|\mathbf{x}) = \frac{P(\mathbf{x})P(\mathbf{y}\|\mathbf{x})}{P(\mathbf{y})}$
Partition function $Z_n(\boldsymbol{J})$	Evidence $P(\mathbf{y})$
Thermal average $\langle f(\boldsymbol{\sigma})\rangle_{\boldsymbol{J}}$	Posterior mean $\mathbb{E}[f(\mathbf{x})\|\mathbf{y}]$
Disorder expectation $\mathbb{E}_{\boldsymbol{J}}$	Data expectation $\mathbb{E}_{\mathbf{y}}$
Free energy $F_n = -\frac{1}{\beta n}\ln Z_n$	Shannon entropy $H(\mathbf{y})/n$, mutual information $I(\mathbf{x};\mathbf{y})/n$.
Control parameter, inverse temperature β	Signal-to-noise ratio λ, relative amount of data n/p
Order parameters, magnetization, overlap	mean-square error, reconstruction error on the signal
Sampling typical/equilibrium configurations	Computing $\hat{\mathbf{x}}_{\text{Bayes}} = \mathbb{E}[\mathbf{x}\|\mathbf{y}]$
Finding the ground state	Computing $\hat{\mathbf{x}}_{\text{MAP}} = \arg\max P(\mathbf{x}\|\mathbf{y})$
Replica-symmetry	Concentration of order parameters
Phase transition	Statistical/computational thresholds
Thermodynamic identity $\frac{\partial}{\partial h}F_n = -\langle m_n\rangle$	I-MMSE formula $\frac{\partial}{\partial\lambda}I(\mathbf{x};\mathbf{y}) = \frac{1}{2}\text{MMSE}(\mathbf{x}\|\mathbf{y})$

- Statistical mechanics provides a powerful framework for inference through a natural mapping between physics and statistics concepts
- Ising/SK models capture essential features of complex systems, and prototypical inference problems are planted versions of statistical mechanics models
- Fundamental limits and threshold phenomena in high-dimensional inference are related to phase transitions

6 (Lect. 9, 10 and 11) Advanced mean-field methods, statics: The replica method

The replica method has been developed for the study of the thermodynamic properties of disordered statistical mechanical models such as spin glasses. It is based on one of the following equivalent identities, coined *replica trick*:

$$\mathbb{E}\ln\mathcal{Z}(\mathbf{Y}) = \lim_{k\to 0_+}\frac{\mathbb{E}[\mathcal{Z}(\mathbf{Y})^k]-1}{k} = \lim_{k\to 0_+}\frac{\partial}{\partial k}\ln\mathbb{E}[\mathcal{Z}(\mathbf{Y})^k] = \lim_{k\to 0_+}\frac{\ln\mathbb{E}[\mathcal{Z}(\mathbf{Y})^k]}{k}$$

where the *replicated partition function* is

$$\mathcal{Z}(\mathbf{y})^k := \int dP(\{\mathbf{x}_1^k\})\exp\Big\{-\sum_{a=1}^{k}\mathcal{H}(\mathbf{x}^a;\mathbf{y})\Big\}$$

using the notation $\int dP(\{\mathbf{x}_1^k\})\cdots = \int_{\mathbb{R}^{nk}}\prod_{a=1}^{k}\prod_{i=1}^{n}dP_X(x_i^a)\cdots$, and where the Hamiltonian is (32). This is the partition function of k replicas $\{\mathbb{R}^n \ni \mathbf{x}^a := \mathbf{x}^{(a)} : a = 1,\dots,k\}$ drawn i.i.d. from the posterior (31) which is $\propto \exp\{-\mathcal{H}(\cdot\,;\mathbf{y})\}$. Choosing your favorite form of the replica trick, the asymptotic free energy (related to the mutual information by an additive constant) reads

$$\lim_{n\to+\infty} f_n = -\lim_{n\to+\infty}\frac{1}{n}\mathbb{E}\ln\mathcal{Z}(\mathbf{Y}) = -\lim_{n\to+\infty}\lim_{k\to 0_+}\frac{\ln\mathbb{E}[\mathcal{Z}(\mathbf{Y})^k]}{nk}. \tag{35}$$

We therefore compute the moments of the partition function $\mathbb{E}[\mathcal{Z}(\mathbf{Y})^k]$ as if $k \in \mathbb{N}$ and then we'll do an analytic continuation to the reals with $k \to 0_+$ and hope for the best.

We compute

$$\mathbb{E}[\mathcal{Z}(\mathbf{Y})^k] = \mathbb{E}_{\mathbf{X}^*}\mathbb{E}_{\mathbf{Y}|\mathbf{X}^*} \int dP(\{\mathbf{x}_1^k\}) \exp \sum_{i<j} \Big(y_{ij} \sqrt{\frac{\lambda}{n}} \sum_{a=1}^{k} x_i^a x_j^a - \frac{\lambda}{2n} \sum_{a=1}^{k} (x_i^a x_j^a)^2 \Big).$$

We now integrate the quenched Gaussian variables

$$y_{ij} \sim \mathcal{N}\Big(\sqrt{\frac{\lambda}{n}} X_i^* X_j^*, 1 \Big).$$

Using the Gaussian integration formula (also called Hubbard-Stratonovich formula when seen from right to left)

$$\int_{\mathbb{R}} dz \exp\{-az^2 + bz\} = \sqrt{\frac{\pi}{a}} \exp \frac{b^2}{4a}$$

and, letting the sum $\sum_{a\neq b}^{u,k}$ be over $\{a, b \in \{u, u+1, \dots, k\} : a \neq b\}$, we obtain

$$\mathbb{E}[\mathcal{Z}(\mathbf{Y})^k] = \mathbb{E}_{\mathbf{X}^*} \int dP(\{\mathbf{x}_1^k\}) \exp \sum_{i<j} \Big\{ \frac{\lambda}{n} \sum_{a=1}^{k} x_i^a X_i^* x_j^a X_j^* + \frac{\lambda}{2n} \sum_{a\neq b}^{1,k} x_i^a x_i^b x_j^a x_j^b \Big\}.$$

Because the prior matches the ground-truth distribution $\mathbb{E}_{\mathbf{X}^*} \cdots = \int dP(\mathbf{x}^*) \cdots$. Then, denoting $\mathbf{x}^* = \mathbf{x}^0$ and $\int dP(\{\mathbf{x}_0^k\}) \cdots = \int_{\mathbb{R}^{n(k+1)}} \prod_{a=0}^{k} \prod_{i=1}^{n} dP_X(x_i^a) \cdots$, the replicated partition function can be re-expressed as

$$\begin{aligned}
\mathbb{E}[\mathcal{Z}(\mathbf{Y})^k] &= \int dP(\{\mathbf{x}_0^k\}) \exp \frac{\lambda}{2n} \sum_{a\neq b}^{0,k} \sum_{i<j} x_i^a x_i^b x_j^a x_j^b \\
&= \int dP(\{\mathbf{x}_0^k\}) \exp \frac{\lambda}{4n} \sum_{a\neq b}^{0,k} \Big\{ \Big(\sum_{i=1}^{n} x_i^a x_i^b \Big)^2 - \sum_{i=1}^{n} (x_i^a x_i^b)^2 \Big\}. \qquad (36)
\end{aligned}$$

We observe here a direct consequence of the fact that we are considering the problem in the Bayesian optimal setting (i.e., the posterior is known): the ground-truth plays a perfectly symmetric role as a replica (i.e., a sample from the posterior), thus the notation $\mathbf{x}^* = \mathbf{x}^0$. Averaging the (random) quenched disorder $\mathbf{Y}$ therefore converted a disordered system made of k independent copies/replicas, each dependent on $\mathbf{Y}$, into a non-disordered system but where the replicas are now coupled.

This identity shows that the replicated partition function depends on the "microscopic configurations" $\{\mathbf{x}^a\}$ only through the "macroscopic" overlap order parameter $Q_{ab} = Q(\mathbf{x}^a, \mathbf{x}^b) := \frac{1}{n} \mathbf{x}^a \cdot \mathbf{x}^b$:

$$\mathbb{E}[\mathcal{Z}(\mathbf{Y})^k] = \int dP(\{\mathbf{x}_0^k\}) \exp \frac{\lambda n}{4} \sum_{a\neq b}^{0,k} Q(\mathbf{x}^a, \mathbf{x}^b)^2.$$

We dropped the sub-dominant $\frac{\lambda}{4n} \sum_{a\neq b}^{0,k} \sum_{i=1}^{n} (x_i^a x_i^b)^2$ from (36) that plays no role in the large n limit. The expression above suggests to introduce the multiplicity $\exp \Gamma(\boldsymbol{q})$ of the configurations $\{\mathbf{x}^a\}$ which have a given overlap matrix $(Q_{ab}) = (q_{ab})$ through the identity:

$$\exp \Gamma(\boldsymbol{q}) := \int dP(\{\mathbf{x}_0^k\}) \prod_{a\neq b}^{0,k} \delta\big(q_{ab} - Q(\mathbf{x}^a, \mathbf{x}^b)\big).$$

So $\Gamma(\boldsymbol{q})$ is interpreted as the entropy associated with the configurations with overlap $\boldsymbol{q}$. We therefore have the identity

$$\begin{aligned} 1 &= \int d\boldsymbol{q} \exp \Gamma(\boldsymbol{q}) \\ &= \frac{1}{2\pi} \int_{\mathbb{R}^{nk(k+1)}} dP(\{\mathbf{x}_0^k\}) \int_{\mathbb{R}^{k(k+1)}} d\boldsymbol{q} \int_{\mathrm{i}\mathbb{R}^{k(k+1)}} d\hat{\boldsymbol{q}} \exp \sum_{a\neq b}^{0,k} \hat{q}_{ab}\big(q_{ab} - Q(\mathbf{x}^a, \mathbf{x}^b)\big). \end{aligned} \tag{37}$$

Note that the integral over the $\hat{q}_{ab}$'s are on the imaginary axis. We used

$$1 = \frac{1}{2\pi} \int_{\mathbb{R}} dx \int_{\mathrm{i}\mathbb{R}} d\hat{x} \exp\big\{\hat{x}(x-\bar{x})\big\} = \int dx \delta(x-\bar{x}).$$

This identity is at the origin of the formal Fourier represention of the dirac delta function: $\delta(x) = \frac{1}{2\pi}\int_{\mathrm{i}\mathbb{R}} d\hat{x} \exp(\hat{x}x)$. Note that this can only be used inside an integral over x otherwise the integral $\int_{\mathrm{i}\mathbb{R}} d\hat{x} \exp(\hat{x}x)$ is not convergent[2]. We plug (37) in the replicated partition function:

$$\begin{aligned} &\mathbb{E}[\mathcal{Z}(\mathbf{Y})^k] \\ &= \Big(\frac{1}{2\pi}\Big)^{k(k+1)} \int dP(\{\mathbf{x}_0^k\}) d\boldsymbol{q} d\hat{\boldsymbol{q}} \exp \sum_{a\neq b}^{0,k} \Big\{ \hat{q}_{ab}\Big(\sum_{i=1}^n x_i^a x_i^b - n q_{ab}\Big) + \frac{\lambda n q_{ab}^2}{4} \Big\} \\ &= \Big(\frac{1}{2\pi}\Big)^{k(k+1)} \int d\boldsymbol{q} d\hat{\boldsymbol{q}} \Big(\exp \sum_{a\neq b}^{0,k} \Big\{ \frac{\lambda q_{ab}^2}{4} - \hat{q}_{ab} q_{ab} \Big\} \int_{\mathbb{R}^{k+1}} dP(\mathbf{x}) \exp \sum_{a\neq b}^{0,k} \hat{q}_{ab} x^a x^b \Big)^n \\ &= \int d\boldsymbol{q} d\hat{\boldsymbol{q}} \exp\big\{ - n \mathcal{S}_n(\boldsymbol{q}, \hat{\boldsymbol{q}})\big\} \end{aligned} \tag{38}$$

with effective action

$$\mathcal{S}_n(\boldsymbol{q}, \hat{\boldsymbol{q}}) := \frac{k(k+1)}{n} \ln 2\pi - \sum_{a\neq b}^{0,k} \Big\{ \frac{\lambda q_{ab}^2}{4} - \hat{q}_{ab} q_{ab} \Big\} - \ln \int dP(\mathbf{x}) \exp \sum_{a\neq b}^{0,k} \hat{q}_{ab} x^a x^b.$$

Note that the decoupling in the real space (over the indices i, j) took place automatically as a consequence of the fixed overlap constraint. As $n \to +\infty$ we evaluate this integral by saddle point assuming a replica symmetric ansatz for the solution:

$$q_{ab} = q \quad \text{and} \quad \hat{q}_{ab} = \frac{\hat{q}}{2} \quad \text{for all} \quad a \neq b.$$

Then the saddle point evaluation gives

$$-\lim_{n\to+\infty} \frac{\ln \mathbb{E}[\mathcal{Z}(\mathbf{Y})^k]}{nk} = \operatorname*{extr}_{q,\hat{q}} \frac{\mathcal{S}(q,\hat{q})}{k} \tag{39}$$

where the replica symmetric action is

$$\frac{\mathcal{S}(q,\hat{q})}{k} = (k+1)\Big(\frac{q\hat{q}}{2} - \frac{\lambda q^2}{4}\Big) - \frac{1}{k} \ln \int dP(\mathbf{x}) \exp \frac{\hat{q}}{2} \Big\{ \Big(\sum_{a=0}^k x^a \Big)^2 - \sum_{a=0}^k (x^a)^2 \Big\}.$$

As the integral we estimate by saddle point lives in the complex plane, we need to extremize (not minimize). The extremization (39) is therefore over $q, \hat{q} \in \mathbb{C}$: the effective action is holomorphic so the complex integral (38) is deformed, without changing the integral value, in order to pass through the extremum saddle point.

[2] To cure this issue one can regularize this integral using instead the Poisson kernel $\eta_\varepsilon(x)$ in the limit $\varepsilon \to 0_+$, where $\eta_\varepsilon(x) := \varepsilon/(\pi(\varepsilon^2 + x^2)) = \frac{1}{2\pi}\int_{\mathrm{i}\mathbb{R}} d\hat{x} \exp(\hat{x}x - |\varepsilon\hat{x}|)$

Let $\mathcal{S}(q,\hat{q}) = \lim_{n\to+\infty} \mathcal{S}_n(q,\hat{q})$. In order to decouple the variables in the replica space (indices a, b) we use the Hubbard-Stratonovich transform with $a = 1/2$ and $b = \sqrt{\hat{q}}\sum_{a=0}^{k} x^a$. With $Z \sim \mathcal{N}(0,1)$ it yields

$$\begin{aligned}\frac{\mathcal{S}(q,\hat{q})}{k} &= (k+1)\Big(\frac{q\hat{q}}{2} - \frac{\lambda q^2}{4}\Big) - \frac{1}{k}\ln\int dP(\mathbf{x}) \exp\frac{\hat{q}}{2}\Big\{\Big(\sum_{a=0}^{k} x^a\Big)^2 - \sum_{a=0}^{k}(x^a)^2\Big\} \\ &= (k+1)\Big(\frac{q\hat{q}}{2} - \frac{\lambda q^2}{4}\Big) - \frac{1}{k}\ln\mathbb{E}\int\prod_{a=0}^{k} dP(x^a)\exp\Big\{\sqrt{\hat{q}}Zx^a - \frac{\hat{q}}{2}(x^a)^2\Big\}.\end{aligned}$$

Letting $k \to 0_+$ we obtain that the free energy given by $\lim_{k\to 0_+}\min_{q,\hat{q}} \frac{1}{k}\mathcal{S}(q,\hat{q})$ is (here $X^* \sim P_X$)

$$\lim_{n\to+\infty} f_n = \operatorname*{extr}_{q,\hat{q}}\Big[\frac{q\hat{q}}{2} - \frac{\lambda q^2}{4} - \mathbb{E}\ln\int dP_X(x)\exp\Big\{\sqrt{\hat{q}}Zx + \hat{q}xX^* - \frac{\hat{q}}{2}x^2\Big\}\Big].$$

Let us denote by $f^{(\mathrm{RS})}(q,\hat{q})$ the two-letters replica symmetric potential, i.e., the function above inside the bracket $[\ldots]$ that is being extremized. The stationary conditions for this function are

$$\frac{d}{dq}f^{(\mathrm{RS})}(q,\hat{q}) = 0 \quad \Rightarrow \quad \hat{q} = \lambda q.$$

The second stationary condition gives

$$\begin{aligned}\frac{d}{d\hat{q}}f^{(\mathrm{RS})}(q,\hat{q}) = 0 \quad \Rightarrow \quad q &= 2\,\mathbb{E}\Big\langle \frac{ZX}{2}\frac{1}{\sqrt{\hat{q}}} + XX^* - \frac{X^2}{2}\Big\rangle_{\hat{q}} \\ &= 2\,\mathbb{E}\Big\langle \Big(\frac{X^2}{2} - \frac{XX'}{2}\Big) + XX^* - \frac{X^2}{2}\Big\rangle_{\hat{q}} \\ &\overset{\mathrm{N}}{=} \mathbb{E}\langle XX^*\rangle_{\hat{q}},\end{aligned}$$

where the bracket $\langle - \rangle_{\hat{q}}$ is w.r.t. the measure $\propto dP_X(x)\exp\{\sqrt{\hat{q}}Zx + \hat{q}xX^* - x^2\hat{q}/2\}$. Plugging the first stationnary condition in the two-letters potential, one recovers the single-letter potential that is minimized in the replica formula.